

Computer Networks

A computer network is a set of computers that are connected by cables or wireless devices to share data and resources. A computer network can enable communication among different users, devices, and applications. A computer network can also provide access to remote services and resources, such as the internet, cloud computing, or distributed databases.

Some of the main concepts and components of computer networks are:

- **Network nodes:** These are the devices that participate in the network, such as computers, routers, switches, servers, printers, etc. Network nodes have unique identifiers, such as IP addresses or MAC addresses, that allow them to communicate with each other.
- **Network links:** These are the physical or logical connections that enable data transmission between network nodes, such as wires, Ethernet cables, fibre optics, Wi-Fi, radio waves, etc. Network links have different characteristics, such as bandwidth, latency, reliability, etc., that affect the performance of the network.
- **Network protocols:** These are the rules and standards that govern how network nodes communicate and exchange data, such as TCP/IP, HTTP, FTP, SMTP, etc. Network protocols define the format, structure, and sequence of the data packets that are sent and received over the network.
- **Network topologies:** These are the ways that network nodes and links are arranged and connected, such as bus, star, ring, mesh, tree, etc. Network topologies affect the cost, complexity, scalability, and reliability of the network.
- **Network architectures:** These are the models and designs that describe the structure, organization, and functions of the network, such as client-server, peer-to-peer, cloud computing, etc. Network architectures affect the security, efficiency, and flexibility of the network.

Some of the main types and categories of computer networks are:

- **Local-area network (LAN):** This is a network that connects devices in a limited physical area, such as a home, office, or campus. A LAN can be small or large, depending on the number and type of devices and links. A LAN can be wired or wireless, or a combination of both. A LAN can be private or public, depending on the access and security policies. A LAN can use different network protocols, such as Ethernet, Wi-Fi, Bluetooth, etc.
- **Wide-area network (WAN):** This is a network that connects devices across a large geographical area, such as a country, continent, or the world. A WAN can span multiple LANs or other networks, such as the internet. A WAN can use different network links, such as telephone lines, satellite links, optical fibers, etc. A WAN can use different network protocols, such as TCP/IP, MPLS, ATM, etc.
- **Enterprise network:** This is a network that connects the devices and

resources of a large organization, such as a corporation, government, or university. An enterprise network can consist of multiple LANs and WANs, as well as other networks, such as intranets, extranets, virtual private networks (VPNs), etc. An enterprise network can use different network architectures, such as client-server, peer-to-peer, cloud computing, etc. An enterprise network can have different network functions, such as data storage, data processing, data analysis, data security, etc.

- **Service-provider network:** This is a network that provides network services and resources to other networks or users, such as the internet service providers (ISPs), telecommunication companies, cloud service providers, etc. A service-provider network can have different network infrastructures, such as data centers, network access points, network backbone, etc. A service-provider network can have different network technologies, such as optical switching, software-defined networking (SDN), network function virtualization (NFV), etc.

Some of the main benefits and challenges of computer networks are:

- **Benefits:** Computer networks can enable faster, easier, and cheaper communication and data exchange among different users, devices, and applications. Computer networks can also enable access to remote and distributed services and resources, such as the internet, cloud computing, or big data. Computer networks can also enhance the performance, reliability, and security of the network and its components, by using techniques such as load balancing, fault tolerance, encryption, etc.
- **Challenges:** Computer networks can also face various problems and issues, such as network congestion, network failure, network attack, network interference, etc. Computer networks can also require high cost, complexity, and maintenance, depending on the size, type, and design of the network. Computer networks can also pose ethical, legal, and social implications, such as privacy, security, censorship,

Unit 1 - Introductory Concepts of Computer Networks and Physical Layer

- A computer network is any group of interconnected computing devices capable of sending or receiving data.
- A computing device isn't just a computer—it's any device that can run a program, such as a tablet, phone, or smart sensor.
- Computer networks can be classified based on their size, topology, architecture, and protocols.
- Some common types of computer networks are:
 - Local Area Network (LAN): A network that connects devices within a small geographic area, such as a home, office, or school.
 - Wide Area Network (WAN): A network that connects devices across a large geographic area, such as a city, country, or the world.

- Metropolitan Area Network (MAN): A network that connects devices within a metropolitan area, such as a city or a campus.
- Personal Area Network (PAN): A network that connects devices within a personal range, such as a Bluetooth headset or a smart-watch.
- Wireless Network: A network that uses wireless technologies, such as radio waves, infrared, or satellite, to transmit data.
- The Open Systems Interconnection (OSI) model is a conceptual framework that defines the functions and interactions of seven layers of a network system.
- The seven layers of the OSI model are:
 - Physical Layer: The lowest layer of the OSI model that deals with the actual physical connection between the devices, such as the hardware equipment, cabling, wiring, frequencies, pulses, etc .
 - Data Link Layer: The layer that provides reliable transmission of data frames between nodes connected by the physical layer, such as error detection, correction, and flow control.
 - Network Layer: The layer that provides routing and switching of data packets between different networks, such as addressing, congestion control, and fragmentation.
 - Transport Layer: The layer that provides end-to-end communication between applications, such as reliability, segmentation, and multiplexing.
 - Session Layer: The layer that establishes, maintains, and terminates sessions between applications, such as authentication, synchronization, and dialog control.
 - Presentation Layer: The layer that provides data representation and encryption for the applications, such as compression, encryption, and translation.
 - Application Layer: The highest layer of the OSI model that provides services and protocols for the applications, such as HTTP, FTP, SMTP, etc.
- The physical layer is responsible for movements of individual bits from one hop (node) to the next.
- The physical layer coordinates the functions required to carry a bit stream over a physical medium.
- The physical layer deals with the mechanical and electrical specifications of the interface and the transmission medium.
- The physical layer provides its services to the data link layer.
- Some of the functions of the physical layer are:
 - Encoding: The process of converting digital data into electrical signals that can be transmitted over the medium.
 - Modulation: The process of varying the amplitude, frequency, or phase of a carrier signal according to the data signal.
 - Multiplexing: The process of combining multiple data signals into one composite signal that can be transmitted over the medium.

- Transmission Mode: The mode of data transmission over the medium, such as simplex, half-duplex, or full-duplex.
- Transmission Rate: The rate at which data is transmitted over the medium, measured in bits per second (bps).
- Transmission Media: The physical medium that carries the data signal, such as twisted pair, coaxial cable, optical fiber, or wireless.
- Some of the advantages of the physical layer are:
 - It enables data communication between different devices over a network.
 - It supports various types of transmission media and modes.
 - It provides the basic infrastructure for data transmission.
- Some of the disadvantages of the physical layer are:
 - It does not provide any error detection or correction.
 - It does not provide any security or encryption.
 - It does not guarantee the delivery or quality of service of the data.
- Some of the examples of the physical layer are:
 - Ethernet: A standard for wired LANs that uses twisted pair or coax

Introductory Concepts of Computer Networks

- A computer network is a system of interconnected devices that can communicate and share resources with each other.
- The devices in a network can be computers, servers, routers, switches, printers, scanners, cameras, etc.
- The devices are also called nodes or hosts in a network.
- A network can be classified based on its size, topology, architecture, and protocols.
- The size of a network refers to the number and distance of the devices in the network. Some common network sizes are:
 - Local Area Network (LAN): A network that covers a small geographical area, such as a home, office, or building. LANs are usually owned and managed by a single organization or individual.
 - Metropolitan Area Network (MAN): A network that covers a larger geographical area, such as a city or a campus. MANs are usually owned and managed by a consortium of organizations or service providers.
 - Wide Area Network (WAN): A network that covers a very large geographical area, such as a country or a continent. WANs are usually owned and managed by multiple organizations or service providers, and use public or leased communication links.
- The topology of a network refers to the physical or logical arrangement of the devices and the links in the network. Some common network topologies are:
 - Bus: A network where all the devices are connected to a single cable or backbone. The cable acts as a shared medium for data transmission. A bus network is simple and cheap, but has low performance

and reliability.

- Star: A network where all the devices are connected to a central device, such as a hub or a switch. The central device acts as a point of connection and control for data transmission. A star network is easy to install and manage, but has high cost and dependency on the central device.
 - Ring: A network where all the devices are connected to form a closed loop. The data travels in one direction around the loop. A ring network is simple and reliable, but has low scalability and fault tolerance.
 - Mesh: A network where each device is connected to every other device. The data can take multiple paths to reach the destination. A mesh network is robust and resilient, but has high complexity and overhead.
 - Tree: A network where the devices are arranged in a hierarchical structure, such as a parent-child relationship. The data flows from the root node to the leaf nodes, or vice versa. A tree network is scalable and efficient, but has high latency and maintenance.
 - Hybrid: A network that combines two or more of the above topologies. A hybrid network can have the advantages of different topologies, but also the disadvantages and compatibility issues.
- The architecture of a network refers to the design and organization of the network layers, functions, and protocols. Some common network architectures are:
 - Peer-to-peer (P2P): A network where each device can act as both a client and a server, and communicate directly with each other. A P2P network is decentralized and self-organizing, but has low security and quality of service.
 - Client-server: A network where some devices act as servers that provide services or resources to other devices, which act as clients that request or consume the services or resources. A client-server network is centralized and efficient, but has high cost and vulnerability.
 - Cloud: A network where the services or resources are provided by a network of servers that are hosted on the internet, and accessed by the devices through a web browser or an application. A cloud network is scalable and flexible, but has low privacy and control.
 - The protocols of a network refer to the rules and standards that govern the communication and interaction of the devices in the network. Some common network protocols are:
 - Transmission Control Protocol/Internet Protocol (TCP/IP): A suite of protocols that defines the basic functions of the internet, such as addressing, routing, error control, and data transmission. TCP/IP consists of four layers: application, transport, internet, and network access.
 - Hypertext Transfer Protocol (HTTP): A protocol that defines how web browsers and web servers communicate and exchange web pages and other resources. HTTP is based on the request-response model,

where the client sends a request to the server, and the server sends a response to the client.

- File Transfer Protocol (FTP): A protocol that defines how files are transferred between devices over a network. FTP uses two connections: a control connection for exchanging commands and messages, and a data connection for transferring files.
- Simple Mail Transfer Protocol (SMTP): A protocol that defines how email messages are sent and received over a network. SMTP uses a store-and-forward mechanism, where the messages are stored in intermediate servers until they reach the destination.
- Domain Name System (

Goals and applications of networks and protocols

A network is a system of two or more devices that are connected to each other and can communicate and share resources. A protocol is a set of rules and standards that define how devices on a network exchange information and cooperate.

Some of the main goals and applications of networks and protocols are:

- **Resource sharing:** Networks allow devices to share hardware and software resources, such as printers, scanners, storage, applications, etc. This can reduce the cost and increase the efficiency of using these resources. For example, a group of office workers can share a common printer instead of having individual printers for each computer .
- **High reliability:** Networks provide multiple sources of supply and backup for data and services. This can improve the availability and fault-tolerance of the network. For example, if one server fails, another server can take over its tasks and prevent data loss or service interruption .
- **Greater flexibility:** Networks enable devices to connect and communicate with each other regardless of their physical location and type. This can increase the mobility and scalability of the network. For example, a laptop can connect to the internet via a wireless network and access data and services from anywhere .
- **Increased productivity:** Networks facilitate data access and collaboration among users and devices. This can enhance the speed and quality of work and decision making. For example, a team of workers can share and edit documents online and communicate via email or video conferencing .
- **Connection:** Protocols establish and maintain stable connections between different devices on the same network. This can ensure the proper delivery and reception of data and services. For example, the Transmission Control Protocol (TCP) ensures that data packets are sent and received reliably and in order.
- **Link aggregation:** Protocols allow devices to combine multiple network connections into one link between two devices. This can improve the band-

width and redundancy of the network. For example, the Link Aggregation Control Protocol (LACP) enables devices to bundle several physical links into a single logical link.

- **Security:** Protocols provide mechanisms for protecting data and devices from unauthorized access and attacks. This can ensure the confidentiality, integrity, and availability of the network. For example, the Secure Sockets Layer (SSL) protocol encrypts and authenticates data exchanged over the internet.
- **Routing:** Protocols determine the best path for data to travel from one device to another on a network. This can optimize the performance and efficiency of the network. For example, the Open Shortest Path First (OSPF) protocol calculates the shortest and least congested route for data packets.
- **Synchronization:** Protocols coordinate the timing and order of data transmission and reception among devices on a network. This can prevent data loss and corruption due to delays or collisions. For example, the Network Time Protocol (NTP) synchronizes the clocks of devices on a network to a common reference time.

Categories of networks in computer networks

A computer network is a collection of devices that are connected by communication channels to share data and resources. There are different ways to categorize computer networks based on their size, scope, topology, architecture, and purpose. Some of the common categories of networks are:

- **Personal Area Network (PAN):** A PAN is a network that connects devices within a small area, such as a room or a desk. A PAN typically consists of a personal computer, a smartphone, a printer, a scanner, a headset, etc. A PAN can use wired or wireless technologies, such as USB, Bluetooth, infrared, etc. A PAN is used for personal or individual purposes, such as transferring files, printing documents, synchronizing data, etc.
- **Local Area Network (LAN):** A LAN is a network that connects devices within a limited geographical area, such as a building, a campus, or a home. A LAN typically consists of multiple computers, servers, printers, routers, switches, etc. A LAN can use wired or wireless technologies, such as Ethernet, Wi-Fi, etc. A LAN is used for local or organizational purposes, such as sharing files, accessing the internet, hosting applications, etc.
- **Metropolitan Area Network (MAN):** A MAN is a network that connects devices within a large metropolitan area, such as a city or a town. A MAN typically consists of multiple LANs, bridges, routers, etc. A MAN can use wired or wireless technologies, such as fiber optic, microwave, etc. A MAN is used for regional or public purposes, such as providing internet access, connecting government offices, broadcasting media, etc.

- **Wide Area Network (WAN):** A WAN is a network that connects devices across a large geographical area, such as a country or a continent. A WAN typically consists of multiple MANs, routers, gateways, satellites, etc. A WAN can use wired or wireless technologies, such as telephone lines, cable, cellular, etc. A WAN is used for global or commercial purposes, such as connecting different branches of an organization, enabling e-commerce, supporting online gaming, etc.
- **Wireless Local Area Network (WLAN):** A WLAN is a network that connects devices using wireless technologies, such as radio waves, infrared, etc. A WLAN typically consists of wireless access points, wireless routers, wireless adapters, etc. A WLAN can be a subset of a LAN or a MAN, or it can be a standalone network. A WLAN is used for wireless or mobile purposes, such as connecting laptops, tablets, smartphones, etc.
- **Storage Area Network (SAN):** A SAN is a network that connects storage devices, such as hard disks, tape drives, optical disks, etc. A SAN typically consists of servers, switches, controllers, etc. A SAN can use wired or wireless technologies, such as SCSI, Fibre Channel, iSCSI, etc. A SAN is used for storage or backup purposes, such as storing large amounts of data, archiving data, recovering data, etc.
- **Virtual Private Network (VPN):** A VPN is a network that connects devices using a secure and encrypted tunnel over a public network, such as the internet. A VPN typically consists of VPN servers, VPN clients, VPN protocols, etc. A VPN can use wired or wireless technologies, such as SSL, IPSec, PPTP, etc. A VPN is used for secure or remote purposes, such as accessing a private network, protecting data, bypassing censorship, etc.

Some possible mnemonics and learning tricks for the categories of networks are:

- **PAN** can be remembered as **P**ersonal **A**rea **N**etwork or **P**hone **A**nd **N**otebook.
- **LAN** can be remembered as **L**ocal **A**rea **N**etwork or **L**arge **A**partment **N**etwork.
- **MAN** can be remembered as **M**etropolitan **A**rea **N**etwork or **M**assive **A**rea **N**etwork.
- **WAN** can be remembered as **W**ide **A**rea **N**etwork or **W**orldwide **A**rea **N**etwork.
- **WLAN** can be remembered as **W**ireless **L**ocal **A**rea **N**etwork or **W**i-Fi **L**AN.
- **SAN** can be remembered as **S**torage **A**rea **N**etwork or **S**ave **A**ll **N**ow.
- **VPN** can be remembered as **V**irtual **P**rivate **N**etwork or

Organization of the Internet

- The Internet is a global network of interconnected computers and devices

that communicate using standardized protocols and procedures .

- The Internet is composed of many smaller networks, called internets, that are not connected to the global Internet but use the same standards.
- The Internet is organized at four levels:
 - Computer and network hardware: This level consists of the physical devices and components that enable data transmission, such as routers, switches, cables, modems, etc.
 - Internet access and Internet service providers (ISPs): This level consists of the organizations and entities that provide access to the Internet, such as telecom companies, cable operators, wireless providers, etc.
 - Navigation within the Internet: This level consists of the tools and services that help users find and access information on the Internet, such as domain names, IP addresses, search engines, web browsers, etc.
 - Individual sites and communities: This level consists of the websites, platforms, applications, and online groups that create and share content and services on the Internet, such as social media, e-commerce, e-learning, etc.
- The Internet is governed by a variety of stakeholders, such as governments, intergovernmental organizations, technical bodies, civil society, academia, and users, who collaborate and coordinate through various forums and mechanisms.
- The Internet is constantly evolving and facing new challenges and opportunities, such as security, privacy, digital inclusion, innovation, and human rights.

: **Internet Organization - freesoft.org** The organization of the Internet - Oxford University Press
Who Makes the Internet Work: The Internet Ecosystem - Internet Society
Build, Promote, and Defend the Internet - Internet Society

ISP

- ISP stands for Internet Service Provider. It is a company or organization that provides access to the internet and other related services such as web hosting, email, and domain name registration.
- ISPs can be classified into different types based on the level of service they offer, the technology they use, and the customers they serve. Some common types of ISPs are:
 - Access ISPs: They provide internet access to end users or customers, usually through a modem, cable, DSL, fiber, wireless, or satellite connection. Examples of access ISPs are Comcast, AT&T, Verizon, and Spectrum.
 - Hosting ISPs: They provide web hosting services, such as storing and serving web pages, email, databases, and applications. Examples of

hosting ISPs are GoDaddy, Bluehost, HostGator, and Amazon Web Services.

- Transit ISPs: They provide network access to other ISPs, usually by charging a fee based on the amount of traffic they carry. Examples of transit ISPs are Level 3, Cogent, GTT, and NTT.
- Virtual ISPs: They provide internet services by reselling or leasing the network infrastructure and bandwidth of another ISP. Examples of virtual ISPs are NetZero, EarthLink, and AOL.
- Wireless ISPs: They provide internet access using wireless technologies, such as Wi-Fi, WiMAX, LTE, or 5G. Examples of wireless ISPs are T-Mobile, Sprint, and MetroPCS.
- ISPs are connected to each other through network access points (NAPs), internet exchange points (IXPs), or peering agreements. These are locations where ISPs exchange traffic and data with each other, either for free or for a fee.
- ISPs are regulated by different laws and policies in different countries and regions. Some of the issues that affect ISPs are net neutrality, privacy, censorship, security, and taxation.

Network structure with reference to Computer Networks

- Network structure is the way network devices and services are organized and connected to serve the communication needs of client devices.
- Network structure consists of two main components: network architecture and network topology.
- Network architecture defines the physical and logical framework of a computer network, including the hardware, software, transmission media, and communication protocols .
- Network topology describes the shape and layout of a network, such as how the nodes and links are arranged and connected.
- There are different types and levels of network architecture and topology, depending on the scale, scope, and purpose of the network.
- Some of the common network architectures are:
 - LAN (Local Area Network): A network that connects computers over a relatively short distance, such as within a building or a campus.
 - WLAN (Wireless Local Area Network): A network that uses wireless technology, such as Wi-Fi, to connect devices without cables.
 - MAN (Metropolitan Area Network): A network that covers a larger area than a LAN, such as a city or a region.
 - WAN (Wide Area Network): A network that spans a large geographic area, such as a country or a continent, and connects multiple LANs or MANs.
 - PAN (Personal Area Network): A network that connects personal devices, such as smartphones, laptops, or wearable devices, over a short range.
 - VPN (Virtual Private Network): A network that creates a secure and

encrypted connection over a public network, such as the Internet, to access private resources, such as a corporate network.

- Some of the common network topologies are:
 - Bus: A network where all the nodes are connected to a single cable or backbone.
 - Star: A network where all the nodes are connected to a central hub or switch.
 - Ring: A network where the nodes are connected in a circular loop.
 - Mesh: A network where the nodes are connected to each other in a full or partial mesh, meaning that each node has a direct or indirect link to every other node.
 - Tree: A network where the nodes are arranged in a hierarchical structure, with a root node at the top and branches and leaves below.
 - Hybrid: A network that combines two or more of the above topologies.
- One of the most widely used network models is the TCP/IP model, which defines four layers of communication for network devices:
 - Network access layer: Defines how the data gets physically transferred over the transmission media, such as cables or wireless signals.
 - Internet layer: Packages the data into understandable packets so it can be sent and received over the network, using protocols such as IP (Internet Protocol).
 - Transport layer: Allows the network devices to maintain conversations and ensure reliable and error-free data delivery, using protocols such as TCP (Transmission Control Protocol) or UDP (User Datagram Protocol).
 - Application layer: Establishes how high-level applications access the network for purposes of data transfer, using protocols such as HTTP (Hypertext Transfer Protocol) or FTP (File Transfer Protocol).
- Network structure is important for computer networks because it determines the performance, reliability, security, scalability, and functionality of the network. A well-designed network structure can optimize the network resources, reduce the network congestion and latency, enhance the network security and privacy, and support the network growth and evolution.

Network architecture with reference to Computer Networks

- Network architecture is the design of a computer network .
- It is a framework for the specification of a network's physical components and their functional organization and configuration, its operational principles and procedures, as well as communication protocols used .
- Network architecture components include hardware, software, transmission media (wired or wireless), network topology, and communications protocols .
- Based on the network's size and purpose, numerous ways to design network

architecture exist .

- Here are the most common and widely used computer network types :
 - LAN (local area network): A LAN connects computers over a relatively short distance, allowing them to share data, files, printers, and other resources. A LAN typically covers a single building or a group of buildings, such as a home, office, or campus. A LAN can use Ethernet, Wi-Fi, or other technologies to connect devices.
 - WLAN (wireless local area network): A WLAN is just like a LAN but connections between devices on the network are wireless, using radio waves or infrared signals. A WLAN can provide mobility, flexibility, and scalability to a LAN. A WLAN typically uses Wi-Fi or Bluetooth to connect devices.
 - WAN (wide area network): A WAN connects computers over a large geographical area, such as a city, country, or the world. A WAN can span multiple LANs and use different technologies to connect them, such as leased lines, satellite links, or the Internet. A WAN can provide access to remote resources, such as web servers, email servers, or cloud services.
 - MAN (metropolitan area network): A MAN connects computers over a metropolitan area, such as a city or a town. A MAN can cover a larger area than a LAN but a smaller area than a WAN. A MAN can use fiber-optic cables, microwave links, or other technologies to connect devices. A MAN can provide high-speed data transfer and support multimedia applications, such as video conferencing or streaming.
 - PAN (personal area network): A PAN connects devices around a single person, such as a laptop, smartphone, smartwatch, or headphones. A PAN can enable data exchange, synchronization, and communication between personal devices. A PAN typically uses Bluetooth, NFC, or USB to connect devices.
 - SAN (storage area network): A SAN connects storage devices, such as hard disks, tape drives, or optical disks, to servers or other computers. A SAN can provide high-performance, reliable, and secure data storage and backup. A SAN can use SCSI, Fibre Channel, or iSCSI to connect devices.
 - CAN (campus area network): A CAN connects computers over a campus area, such as a university, school, or research center. A CAN can consist of multiple LANs and use switches, routers, or firewalls to connect them. A CAN can provide access to academic resources, such as libraries, laboratories, or classrooms.
 - VPN (virtual private network): A VPN connects computers over a public network, such as the Internet, but creates a secure and private tunnel for data transmission. A VPN can encrypt, authenticate, and compress data to ensure its confidentiality, integrity, and efficiency. A VPN can use IPsec, SSL, or PPTP to connect devices.
 - BAN (body area network): A BAN connects devices on or inside a

human body, such as sensors, implants, or wearable devices. A BAN can monitor, diagnose, or treat various health conditions, such as heart rate, blood pressure, or glucose level. A BAN can use ZigBee, Bluetooth, or RFID to connect devices.

- A mnemonic to remember the network types is **LWMPSCVB** (pronounced as “lum-pix-cub”), where each letter stands for the first letter of each network type. For example, L stands for LAN, W stands for WAN, and so on.
- A learning trick to understand the network types is to compare them with different modes of transportation, such as cars, buses, trains, planes, bikes, etc. For example, a LAN is like a car that can transport a few people within a short distance, a WAN is like a plane that can transport many people across a long distance, a PAN is like a bike

Layering principles with reference to Network Architecture in Computer Networks

- Layering is a mechanism that divides a complex system into smaller and manageable parts, called layers, that can communicate with each other through well-defined interfaces.
- Layering allows for modularity, abstraction, encapsulation, and interoperability of network components and protocols.
- Layering also facilitates the design, implementation, testing, and maintenance of network systems, as well as the development of new standards and technologies.
- One of the most widely used models of layered network architecture is the Open Systems Interconnection (OSI) model, which consists of seven layers: physical, data link, network, transport, session, presentation, and application .
- The principles that were applied to arrive at the seven layers of the OSI model can be briefly summarized as follows:
 - A layer should be created where a different abstraction is needed.
 - Each layer should perform a well-defined function.
 - The function of each layer should be chosen with an eye toward defining internationally standardized protocols.
 - The layer boundaries should be chosen to minimize the information flow across the interfaces.
 - The number of layers should be large enough that distinct functions need not be thrown together in the same layer out of necessity, and small enough that the architecture does not become unwieldy.
- The main functions and examples of protocols for each layer of the OSI model are:
 - Physical: This layer defines the electrical, mechanical, and procedural characteristics of the physical medium that carries the data signals. Examples of physical layer protocols are Ethernet, RS-232, and USB.
 - Data link: This layer provides reliable and error-free transmission

- of data frames between adjacent nodes on the same physical link. Examples of data link layer protocols are HDLC, PPP, and MAC.
- Network: This layer provides routing and addressing of data packets across different networks or subnets. Examples of network layer protocols are IP, ICMP, and ARP.
- Transport: This layer ensures complete and reliable delivery of data segments between end-to-end applications. Examples of transport layer protocols are TCP, UDP, and SCTP.
- Session: This layer establishes, maintains, and terminates sessions between applications on different hosts. Examples of session layer protocols are RPC, NFS, and SMB.
- Presentation: This layer provides data representation, encryption, compression, and translation services for the application layer. Examples of presentation layer protocols are SSL, TLS, and MIME.
- Application: This layer provides the interface and functionality for user applications and network services. Examples of application layer protocols are HTTP, FTP, and SMTP.
- A possible mnemonic to remember the seven layers of the OSI model from bottom to top is “Please Do Not Throw Sausage Pizza Away” or “All People Seem To Need Data Processing”.

Services in networks architecture in Computer Networks

- Services are the **capabilities** that each layer in a network architecture provides to the layer above it.
- Services are **implemented** by using protocols, which are sets of rules and formats that define the communication between entities in a network.
- Services can be **classified** into two types: connection-oriented and connectionless.
- Connection-oriented services **require** the establishment of a logical connection between the sender and the receiver before any data can be transferred. They provide **reliable, ordered, and error-free** delivery of data. An example of a connection-oriented service is TCP.
- Connectionless services **do not require** any connection setup or teardown. They provide **unreliable, unordered, and error-prone** delivery of data. An example of a connectionless service is UDP.
- Services can also be **categorized** into three levels: application, network, and data link.
- Application services are the **highest-level** services that provide functionality to end-user applications, such as web browsing, email, file transfer, etc. They are often implemented using a **client-server** or **peer-to-peer** architecture based on application layer network protocols. Examples of application services are HTTP, SMTP, FTP, etc.
- Network services are the **intermediate-level** services that provide connectivity and routing between different networks or subnetworks. They are responsible for **addressing, forwarding, and delivering** packets across

the network. They are often implemented using **routers** and **switches**. Examples of network services are IP, ICMP, ARP, etc.

- Data link services are the **lowest-level** services that provide physical transmission of bits over a single link or medium. They are responsible for **framing**, **error detection**, and **flow control** of data. They are often implemented using **network interface cards** and **cables**. Examples of data link services are Ethernet, Wi-Fi, Bluetooth, etc.

protocols and standards in networks architecture in Computer Networks

- Network protocols are a set of guidelines governing the exchange of information in a simple, dependable and secure way.
- Network protocols define how devices communicate within a network, including the formats, procedures, and rules of data transmission.
- Network standards are formal specifications that ensure compatibility and interoperability among different network devices, systems, and applications.
- Network standards are usually developed by standard organizations, such as IEEE, ISO, IETF, etc., or by industry consortia, such as W3C, Bluetooth SIG, etc.
- Network architecture is the design and structure of a network, including its components, connections, and functions.
- Network architecture components include hardware, software, transmission media (wired or wireless), network topology, and communications protocols.
- Network architecture can be classified into two main types: peer-to-peer (P2P) and client/server.
- In a P2P network, each node can act as both a client and a server, and there is no central authority or server.
- In a client/server network, one or more servers provide services to clients, and there is a clear distinction between the roles of each node.
- Network protocols and standards are usually organized into layers, which abstract the functionality and complexity of each level.
- The most common layered models are the OSI model and the TCP/IP model, which have seven and four layers respectively.
- The OSI model consists of the following layers: physical, data link, network, transport, session, presentation, and application.
- The TCP/IP model consists of the following layers: network access, internet, transport, and application.
- Each layer in a model uses protocols and standards to communicate with the same layer in another node, and to provide services to the upper and lower layers.
- Some of the common network protocols and their functions are:
 - Transmission Control Protocol (TCP): provides reliable, ordered, and error-checked delivery of data between applications

- Internet Protocol (IP): provides addressing and routing of data packets across networks
- User Datagram Protocol (UDP): provides fast and efficient delivery of data without error-checking or ordering
- Hypertext Transfer Protocol (HTTP): provides the rules for web browsers and servers to exchange web pages and other resources
- File Transfer Protocol (FTP): provides the rules for transferring files between computers over a network
- Simple Mail Transfer Protocol (SMTP): provides the rules for sending and receiving email messages
- Domain Name System (DNS): provides the rules for translating domain names into IP addresses
- Secure Sockets Layer (SSL) / Transport Layer Security (TLS): provides the rules for encrypting and authenticating data transmission over a network
- Wireless Application Protocol (WAP): provides the rules for accessing web content and services on mobile devices
- Bluetooth: provides the rules for short-range wireless communication between devices
- Ethernet: provides the rules for data transmission over a wired local area network (LAN)
- Wi-Fi: provides the rules for data transmission over a wireless local area network (LAN)
- A possible mnemonic to remember the OSI model layers is: **P**lease **D**o **N**ot **T**hrow **S**ausage **P**izza **A**way
- A possible mnemonic to remember the TCP/IP model layers is: **N**o **I**nternet **T**oday **A**gain

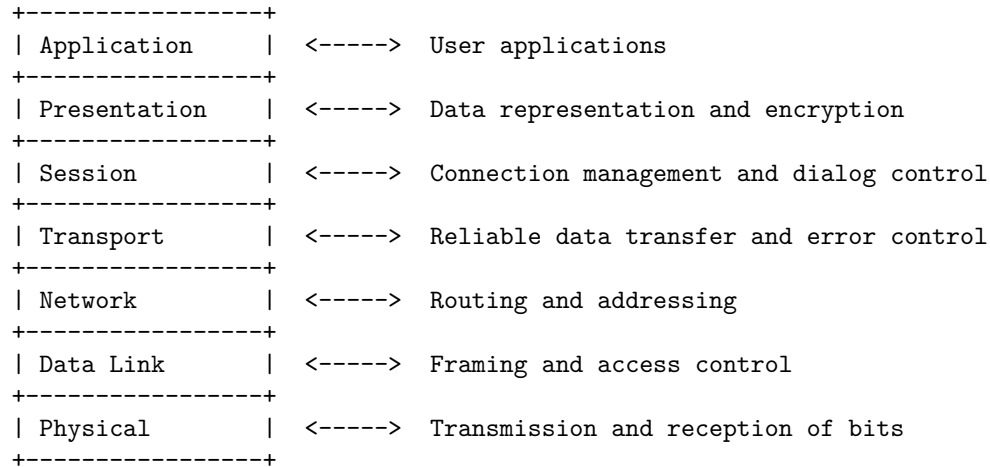
The OSI reference model in in Computer Networks

The OSI reference model is a standard framework that describes how information from a software application in one computer moves through a physical medium to another computer. It was developed by the International Organization for Standardization (ISO) in 1984, and it is now considered as an architectural model for the inter-computer communications .

The OSI reference model divides the whole task of data communication into seven smaller and manageable layers. Each layer is assigned a particular function and interacts with the adjacent layers. The layers are:

- Layer 1: Physical
- Layer 2: Data Link
- Layer 3: Network
- Layer 4: Transport
- Layer 5: Session
- Layer 6: Presentation
- Layer 7: Application

The following diagram shows the OSI reference model and its layers:



The OSI reference model has the following advantages:

- It provides a common language for describing network functions and protocols.
- It facilitates interoperability and compatibility among different network devices and systems.
- It helps in designing, developing, and troubleshooting network applications and protocols.
- It allows modular development and testing of network components and layers.

The OSI reference model has the following disadvantages:

- It is not a protocol itself, but rather a theoretical model that does not reflect the actual implementation of network protocols.
- It is too complex and abstract for some practical purposes, and some layers may be redundant or unnecessary.
- It does not specify the exact services and interfaces for each layer, leaving room for interpretation and variation.

Some mnemonics and learning tricks for remembering the OSI reference model and its layers are:

- All People Seem To Need Data Processing (Application, Presentation, Session, Transport, Network, Data Link, Physical)
- Please Do Not Throw Sausage Pizza Away (Physical, Data Link, Network, Transport, Session, Presentation, Application)
- A Panda Saw Two Naughty Dogs Playing (Application, Presentation, Session, Transport, Network, Data Link, Physical)

TCP/IP protocol suite in Computer Networks

- TCP/IP stands for Transmission Control Protocol/Internet Protocol and is a suite of communication protocols that makes data exchange between two devices possible .
- TCP/IP specifies how data should be packetized, addressed, transmitted, routed, and received on a network by providing end-to-end communication .
- TCP/IP is also used as a communications protocol in a private computer network (an intranet or extranet).
- TCP/IP is commonly known as the Internet protocol suite, as it is the framework for organizing the set of communication protocols used in the Internet and similar computer networks according to functional criteria.
- TCP/IP is based on a four-layer model, which consists of the following layers :
 - Application layer: This layer provides the interface for the user applications to access the network services, such as web browsers, email clients, file transfer programs, etc. The protocols in this layer include HTTP, FTP, SMTP, DNS, etc.
 - Transport layer: This layer provides reliable and/or unreliable data delivery services between the end hosts, such as error detection, flow control, congestion control, etc. The protocols in this layer include TCP, UDP, etc.
 - Internet layer: This layer provides the logical addressing and routing functions for the data packets across the network, such as IP addresses, subnet masks, routers, etc. The protocols in this layer include IP, ICMP, ARP, etc.
 - Network access layer: This layer provides the physical and data link functions for the data transmission over the network medium, such as Ethernet, Wi-Fi, etc. The protocols in this layer include MAC, LLC, etc.
- A mnemonic to remember the four layers of TCP/IP is **A Tea In November**, where the first letter of each word corresponds to the first letter of each layer.
- A diagram to illustrate the TCP/IP protocol suite is shown below:

Application	Application	Application	Ap
Transport	Transport	Transport	T
Internet	Internet	Internet	

Network access	Network access	Network access	Network access
Physical	Physical	Physical	Physical
Network	Network	Network	Network

- Some advantages of TCP/IP protocol suite are :
 - It is widely adopted and supported by various devices and platforms.
 - It is scalable and adaptable to different network sizes and topologies.
 - It is robust and fault-tolerant, as it can handle packet loss, congestion, and errors.
 - It is independent of the underlying network hardware and software, as it provides a common interface for the upper layers.
 - It supports both connection-oriented and connectionless services, as well as both reliable and unreliable data delivery.
- Some disadvantages of TCP/IP protocol suite are :
 - It is complex and difficult to implement and maintain, as it involves many protocols and functions.
 - It is not very secure, as it does not provide encryption or authentication mechanisms by default.
 - It is inefficient, as it adds overhead and redundancy to the data packets.
 - It is not compatible with some older

Network devices in Computer Networks

Network devices are physical devices that enable communication and interaction between hardware on a computer network. Each networking device operates in a distinct computer network segment and performs distinct functions. A network may require hundreds or thousands of different network devices to maintain and build out various LAN and WAN.

Some of the common types of network devices are:

- **Repeater:** A repeater is a device that operates at the physical layer of the OSI model. Its job is to regenerate the signal over the same network, thus extending the transmission distance and reducing the attenuation. Repeaters are useful for connecting segments of a network that use the same media type, such as coaxial cable or fiber optic cable. However, repeaters

cannot filter or modify the data, and they cannot connect segments that use different media types or protocols.

- **Hub:** A hub is a device that operates at the physical layer of the OSI model. It is basically a multi-port repeater that connects multiple wires coming from different branches. A hub broadcasts the data received from one port to all other ports, regardless of the destination address. Hubs are simple and cheap devices, but they create a lot of collisions and waste bandwidth. Hubs are also called concentrators or multi-station access units (MAUs).
- **Bridge:** A bridge is a device that operates at the data link layer of the OSI model. It is used to connect two or more network segments that use the same or different media types, such as Ethernet and Token Ring. A bridge filters the data based on the MAC addresses, and forwards only the frames that belong to the destination segment. Bridges can also learn the MAC addresses of the devices connected to each segment, and create a forwarding table to reduce the broadcast traffic. Bridges can improve the performance and security of a network, but they cannot handle different network architectures or protocols.
- **Switch:** A switch is a device that operates at the data link layer or the network layer of the OSI model. It is similar to a bridge, but with more ports and more intelligence. A switch can create multiple collision domains and reduce the network congestion. A switch can also perform VLANs, which are logical groups of devices that can communicate as if they were on the same physical segment. Switches can also operate at the network layer, and route the data based on the IP addresses. Switches are faster and more scalable than bridges, but they are also more expensive and complex.
- **Router:** A router is a device that operates at the network layer of the OSI model. It is used to connect two or more networks that use the same or different protocols, such as IP, IPX, or AppleTalk. A router forwards the data based on the logical addresses, and can also perform routing protocols, such as RIP, OSPF, or BGP, to find the best path to the destination. Routers can also perform NAT, which is a technique to translate private IP addresses to public IP addresses, and vice versa. Routers can handle different network architectures and protocols, but they are slower and more expensive than switches.
- **Gateway:** A gateway is a device that operates at the application layer of the OSI model. It is used to connect two or more networks that use different protocols, such as TCP/IP and SNA. A gateway converts the data from one format to another, and can also perform protocol conversion, data encryption, or data compression. Gateways are also called protocol converters or application layer gateways. Gateways can enable interoperability between different systems, but they are very complex and costly.

devices.

- **Brouter:** A brouter is a device that combines the functions of a bridge and a router. It operates at both the data link layer and the network layer of the OSI model. It can filter the data based on the MAC addresses or the IP addresses, and can also route the data between different networks. Brouters are also called hybrid devices or routing switches. Brouters can provide both the advantages of bridges and routers, but they are not widely used in practice.
- **NIC:** A NIC, or a network interface card, is a device that operates at the physical layer and the data link layer of the OSI model. It is used to connect a computer or a device to a network, and provides a unique MAC address for identification. A NIC can also perform error detection and correction, and can support different media types and speeds. NICs are also called network adapters or LAN cards. NICs are essential for any network communication, and they are usually integrated into

Network components in Computer Networks

Network components are the hardware and software devices that are used to connect computers and other devices in a network. They enable data transmission, routing, switching, and network management. Some of the basic network components are:

- **Server:** A server is a computer that provides data and services to other computers and users in the network. For example, a web server hosts web pages and responds to requests from web browsers. A server can perform multiple functions, such as file sharing, email, database, printing, etc. A server can be dedicated to one function or be multifunctional.
- **Client:** A client is a computer or device that requests and receives data and services from a server in the network. For example, a web browser is a client that requests and displays web pages from a web server. A client can also be a peer, which means it can act as both a client and a server in the network.
- **Transmission media:** Transmission media are the physical or wireless channels that carry data signals from one device to another in the network. For example, copper wires, fiber optic cables, radio waves, and microwaves are some of the transmission media used in networks. Transmission media can be classified into two types: guided and unguided. Guided media are those that require a physical medium, such as wires or cables. Unguided media are those that do not require a physical medium, such as wireless or satellite.
- **NIC (Network Interface Card):** A NIC is a hardware device that connects a computer or device to the network. It enables the device to send and receive data signals over the transmission media. A NIC has

a unique identifier called a MAC (Media Access Control) address, which is used to identify the device in the network. A NIC can be internal or external, wired or wireless, depending on the type of network and device.

- **Switch:** A switch is a network device that connects multiple devices in a network and forwards data packets to the appropriate destination based on the MAC address of the device. A switch operates at the data link layer of the OSI (Open Systems Interconnection) model, which is responsible for error detection and correction, and media access control. A switch can improve network performance and security by reducing collisions and broadcast traffic.
- **Router:** A router is a network device that connects multiple networks and forwards data packets to the appropriate destination based on the IP (Internet Protocol) address of the device. A router operates at the network layer of the OSI model, which is responsible for addressing, routing, and fragmentation. A router can perform various functions, such as filtering, firewall, NAT (Network Address Translation), DHCP (Dynamic Host Configuration Protocol), etc.
- **Hub:** A hub is a network device that connects multiple devices in a network and broadcasts data packets to all the connected devices. A hub operates at the physical layer of the OSI model, which is responsible for encoding, decoding, and signaling. A hub is a simple and cheap device, but it can cause network congestion and security issues due to collisions and broadcast traffic.
- **Firewall:** A firewall is a network device or software that monitors and controls the incoming and outgoing network traffic based on predefined rules and policies. A firewall can operate at different layers of the OSI model, depending on the type and functionality of the firewall. A firewall can provide network security by preventing unauthorized access, blocking malicious traffic, and logging network activity.
- **Access point:** An access point is a network device that connects wireless devices to a wired network. It acts as a bridge between the wireless and wired networks. An access point can also extend the range and coverage of a wireless network by creating a wireless hotspot. An access point can operate in different modes, such as infrastructure, ad hoc, or mesh.
- **Network software:** Network software are the programs and applications that enable network communication, management, and security. For example, network protocols, such as TCP/IP (Transmission Control Protocol/Internet Protocol), HTTP (Hypertext Transfer Protocol), FTP (File Transfer Protocol), etc., are the rules and standards that govern how data is exchanged and processed in the network. Network operating systems, such as Windows, Linux, MacOS, etc., are the platforms that provide network services and functions, such as user authentication, file sharing,

printing, etc. Network utilities, such as ping, traceroute, ipconfig, etc., are the tools that help diagnose and troubleshoot network problems.

Physical Layer in Computer Networks

- The physical layer is the first and lowest layer of the OSI model of computer networking. It is responsible for sending and receiving bits (0s and 1s) between devices over a physical medium, such as a cable or a wireless channel.
- The physical layer provides an electrical, mechanical, and procedural interface to the transmission medium. It defines how the devices are physically connected, how the bits are represented as signals, and how the signals are transmitted and received.
- The physical layer consists of the electronic circuit transmission technologies of a network, such as twisted pair cables, coaxial cables, fiber-optic cables, radio waves, infrared waves, etc. It can be implemented through a great number of different hardware technologies with widely varying characteristics, such as bandwidth, latency, noise, interference, etc.
- The physical layer performs the following functions:
 - Data rate control: It determines how many bits a sender can send per second over the medium. It depends on the bandwidth and the modulation scheme of the medium.
 - Synchronization of bits: It ensures that the sender and the receiver are synchronized in terms of the timing and the order of the bits. It can be achieved by using clock signals, start and stop bits, or framing techniques.
 - Transmission medium decision: It decides the direction of data transfer over the medium. It can be simplex (one-way), half-duplex (two-way but not at the same time), or full-duplex (two-way at the same time).
 - Physical topology decision: It decides the shape and the arrangement of the devices and the links in the network. It can be mesh, star, bus, ring, etc. It affects the cost, reliability, and performance of the network.
 - Encoding and decoding: It converts the bits into signals that can be transmitted over the medium, and vice versa. It can use different encoding schemes, such as NRZ, RZ, Manchester, etc. It also deals with error detection and correction at the bit level.
 - Multiplexing and demultiplexing: It allows multiple signals to share the same medium, and separates them at the receiver. It can use different multiplexing techniques, such as frequency division multiplexing (FDM), time division multiplexing (TDM), wavelength division multiplexing (WDM), etc.
- A possible mnemonic to remember the functions of the physical layer is **DSTEP ME** (Data rate, Synchronization, Transmission, Encoding, Physical topology, Multiplexing, Error detection).

- A possible learning trick to understand the physical layer is to compare it to a postal service. The physical layer is like the postal workers who deliver the letters (bits) from one address (device) to another over a road (medium). They have to follow certain rules and procedures (interface) to ensure the delivery is successful. They also have to deal with different types of roads (mediums), different speeds and capacities (data rates), different traffic conditions (noise and interference), different directions and routes (transmission and topology), different ways of writing and reading the letters (encoding and decoding), and different ways of sorting and distributing the letters (multiplexing and demultiplexing).

Network topology design in Computer Networks

- Network topology is the arrangement of network devices and the connections between them.
- It defines both the physical and logical aspects of the network, such as the layout of cables, devices, and protocols.
- Network topology affects the performance, reliability, security, and scalability of the network.
- There are two main types of network topology: physical and logical.
 - Physical topology refers to the actual placement and wiring of the network devices and cables.
 - Logical topology refers to the way data flows and communicates across the network, regardless of the physical layout.
- There are several common network topologies, each with its own advantages and disadvantages. Some of the most widely used ones are:
 - Bus topology: A linear network where all devices are connected to a single cable, called the bus. Data is transmitted in both directions along the bus, and each device has a unique address to identify itself.
 - * Advantages: Simple and cheap to install and maintain, easy to add or remove devices.
 - * Disadvantages: Low bandwidth and performance, vulnerable to cable faults and collisions, difficult to troubleshoot and scale.
 - Ring topology: A circular network where each device is connected to two other devices, forming a loop. Data is transmitted in one direction along the ring, and each device acts as a repeater to pass the data to the next device.
 - * Advantages: High bandwidth and performance, no collisions, easy to troubleshoot and monitor.
 - * Disadvantages: Expensive and complex to install and maintain, vulnerable to single point of failure, difficult to add or remove devices.
 - Star topology: A network where each device is connected to a central device, called the hub or switch. Data is transmitted from the source device to the hub, and then to the destination device. The hub acts as a controller and a distributor of the data.

- * Advantages: High bandwidth and performance, robust and reliable, easy to add or remove devices, easy to troubleshoot and scale.
- * Disadvantages: Expensive and complex to install and maintain, dependent on the hub, vulnerable to hub failure.
- Mesh topology: A network where each device is connected to every other device, forming a complete or partial mesh. Data is transmitted along the shortest or optimal path between the source and destination devices, using routing algorithms. The devices act as routers and forwarders of the data.
 - * Advantages: High bandwidth and performance, fault-tolerant and resilient, no single point of failure, easy to scale.
 - * Disadvantages: Very expensive and complex to install and maintain, high cabling and hardware requirements, difficult to troubleshoot and manage.
- Tree topology: A hierarchical network where devices are connected to a root device, which is connected to other devices, forming a tree-like structure. Data is transmitted from the source device to the root device, and then to the destination device, following the branches of the tree. The root device acts as a gateway and a coordinator of the data.
 - * Advantages: High bandwidth and performance, scalable and flexible, easy to add or remove devices, easy to segment and isolate network traffic.
 - * Disadvantages: Expensive and complex to install and maintain, dependent on the root device, vulnerable to root device failure.
- A network topology diagram is a graphical representation of the network topology, showing the devices, connections, and data flow. It helps to visualize and understand the network structure and design.
- A network topology diagram can be created using various tools, such as Microsoft Visio, Lucidchart, Draw.io, etc.
- A network topology diagram should include the following elements:
 - The network devices, such as computers, servers, routers, switches, hubs, etc., labeled with their names, addresses, and functions.
 - The network connections, such as cables, wireless links, etc., labeled with their types, speeds, and directions.
 - The network protocols, such as TCP/IP, Ethernet, etc., labeled with their names, versions, and configurations.
 - The network data flow, such as packets, frames, etc., labeled with their sources, destinations, and contents.
- A network topology diagram should follow the following best practices:
 - Use standard symbols and shapes to represent the network devices and connections, such as circles, squares, lines, etc.
 - Use different colors and styles to distinguish the network devices and connections, such as solid, dashed, dotted, etc.
 - Use consistent and clear labels and annotations to describe the net-

work devices and connections, such as text, numbers, arrows, etc.

Types of connections in Computer Networks

- A connection in a computer network is a link between two or more devices that enables them to communicate and exchange data.
- There are two main types of connections in computer networks: **point-to-point** and **point-to-multipoint**.
- A point-to-point connection is a direct link between two devices, such as a cable, a wireless link, or a circuit-switched connection. A point-to-point connection can be either **simplex**, **half-duplex**, or **full-duplex**.
 - A simplex connection allows data to flow in only one direction, such as a keyboard to a computer or a microphone to a speaker.
 - A half-duplex connection allows data to flow in both directions, but not at the same time, such as a walkie-talkie or a telephone.
 - A full-duplex connection allows data to flow in both directions simultaneously, such as a telephone or a LAN cable.
- A point-to-multipoint connection is a link between one device and multiple devices, such as a broadcast, a multicast, or a multiplexed connection. A point-to-multipoint connection can be either **one-to-many** or **many-to-many**.
 - A one-to-many connection allows data to flow from one device to multiple devices, such as a radio or a TV broadcast, or a hub in a star topology.
 - A many-to-many connection allows data to flow from multiple devices to multiple devices, such as a video conference or a chat room, or a switch in a mesh topology.
- A mnemonic to remember the types of connections is **PHF-BMM** (Point-to-point, Half-duplex, Full-duplex, Broadcast, Multicast, Multiplexed).
- A diagram to illustrate the types of connections is:

Point-to-point

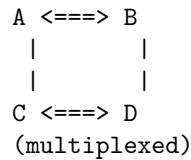
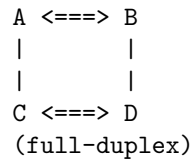
```
A ----- B
|         |
|         |
C ----- D
(simplex)
```

```
A <-----> B
|         |
|         |
C <-----> D
(half-duplex)
```

Point-to-multipoint

```
A ----- B
|         |
|         |
C ----- D
(one-to-many)
```

```
A <-----> B
|         |
|         |
C <-----> D
(many-to-many)
```



- Some advantages of point-to-point connections are:
 - They are simple and easy to set up and maintain.
 - They provide dedicated and secure communication channels.
 - They have high performance and reliability.
- Some disadvantages of point-to-point connections are:
 - They are costly and require more resources and devices.
 - They are not scalable and flexible for large networks.
 - They can create congestion and bottlenecks in the network.
- Some advantages of point-to-multipoint connections are:
 - They are economical and require less resources and devices.
 - They are scalable and flexible for large networks.
 - They can support multiple applications and services.
- Some disadvantages of point-to-multipoint connections are:
 - They are complex and difficult to set up and maintain.
 - They provide shared and less secure communication channels.
 - They have lower performance and reliability.

Transmission media in Computer Networks

Transmission media in computer networks are the physical channels that carry data from one device to another. They can be classified into two major categories: guided and unguided media.

- Guided media are those that provide a physical path for the data to travel, such as wires, cables, and optical fibers. They can be further divided into three types: twisted pair, coaxial, and fiber optic cables.
 - Twisted pair cables consist of two insulated copper wires twisted together to reduce electromagnetic interference. They are commonly used for telephone and Ethernet connections. They have low cost, easy installation, and moderate bandwidth, but they are also susceptible to noise, attenuation, and eavesdropping. A mnemonic to remember the advantages and disadvantages of twisted pair cables is **TAME**: Twisted, Affordable, Moderate, Eavesdroppable.
 - Coaxial cables consist of a central copper core surrounded by an insulating layer, a braided metal shield, and an outer cover. They are used for cable TV, broadband internet, and long-distance telephone networks. They have high bandwidth, low noise, and high security, but they are also expensive, bulky, and difficult to install. A mnemonic to remember the advantages and disadvantages of coaxial cables is **CHUBS**: Coaxial, High, Unwieldy, Bulky, Secure.
 - Fiber optic cables consist of thin glass or plastic fibers that transmit

data using light pulses. They are used for high-speed internet, long-distance communication, and medical imaging. They have very high bandwidth, low attenuation, low interference, and high security, but they are also very expensive, fragile, and require special devices to connect. A mnemonic to remember the advantages and disadvantages of fiber optic cables is **FLASH**: Fiber, Light, Attenuation, Speed, High-cost.

- Unguided media are those that do not provide a physical path for the data to travel, but rather use electromagnetic waves or signals to transmit data through air, space, or water. They can be further divided into three types: radio waves, microwaves, and infrared waves.
 - Radio waves are electromagnetic waves that can travel long distances and penetrate through walls and buildings. They are used for radio, TV, cellular phones, and wireless LANs. They have low cost, easy installation, and wide coverage, but they are also prone to interference, noise, and security issues. A mnemonic to remember the advantages and disadvantages of radio waves is **RICE**: Radio, Inexpensive, Coverage, EMI.
 - Microwaves are electromagnetic waves that have higher frequencies and shorter wavelengths than radio waves. They are used for satellite communication, radar, and wireless WANs. They have high bandwidth, low attenuation, and high security, but they are also expensive, require line-of-sight, and suffer from rain fade. A mnemonic to remember the advantages and disadvantages of microwaves is **MARS**: Microwave, Attenuation, Rain, Sight.
 - Infrared waves are electromagnetic waves that have higher frequencies and shorter wavelengths than microwaves. They are used for remote controls, wireless keyboards and mice, and short-range communication. They have low cost, low interference, and high security, but they are also limited by distance, line-of-sight, and ambient light. A mnemonic to remember the advantages and disadvantages of infrared waves is **ILLS**: Infrared, Light, Line, Short.

Here is a table that summarizes the main characteristics of the different types of transmission media:

Type	Bandwidth	Cost	Installation	Attenuation	Interference	Security	Distance	Line-of-sight
Twisted pair	Moderate	Low	Easy	High	High	Low	Short	No
Coaxial	High	High	Difficult	Low	Low	High	Medium	No
Fiber optic	Very high	Very high	Difficult	Very low	Very low	Very high	Long	No
Radio	Low	Low	Easy	Low	High	Low	Long	No
Microwave	High	High	Difficult	Low	Low	High	Long	Yes
Infrared	Low	Low	Easy	High	Low	High	Short	Yes

Signal transmission and encoding in Computer Networks

- Signal transmission is the process of sending digital or analog data over a physical medium such as copper wire, optical fiber, or wireless channel.
- Encoding is the process of converting data into a specific format that can be recognized and interpreted by the receiver.
- Encoding can be classified into two types: digital-to-digital encoding and analog-to-digital encoding.
- Digital-to-digital encoding is the representation of digital information by a digital signal. It is also known as line coding. It involves converting a stream of bits into a series of voltage pulses, light pulses, or electromagnetic waves that can be transmitted over a wire or a wireless channel.
- There are different techniques of digital-to-digital encoding, such as:
 - Non-return-to-zero (NRZ): In this technique, a high voltage represents a 1 bit and a low voltage represents a 0 bit. There is no change in the signal level for consecutive bits of the same value. This technique is simple but suffers from synchronization problems and DC component issues.
 - Return-to-zero (RZ): In this technique, a high voltage represents a 1 bit and a low voltage represents a 0 bit. However, the signal level returns to zero halfway through each bit duration. This technique reduces the DC component problem but increases the bandwidth requirement and the synchronization problem.
 - Manchester: In this technique, a transition from low to high voltage represents a 1 bit and a transition from high to low voltage represents a 0 bit. There is always a transition at the middle of each bit duration. This technique eliminates the DC component problem and the synchronization problem but doubles the bandwidth requirement.
 - Differential Manchester: In this technique, a transition at the beginning of a bit duration represents a 0 bit and no transition at the beginning of a bit duration represents a 1 bit. There is always a transition at the middle of each bit duration. This technique is similar to Manchester but uses differential encoding, which means that the signal is encoded based on the previous bit rather than the current bit.
 - 4B/5B: In this technique, every 4 bits of data are encoded into 5 bits of signal, using a predefined mapping table. The 5-bit signal has at least one transition and no more than two consecutive 0s or 1s. This technique reduces the DC component problem and the synchronization problem but increases the overhead by 25%.
- Analog-to-digital encoding is the representation of analog information by a digital signal. It is also known as digitization or sampling. It involves converting a continuous analog signal into a discrete sequence of bits that

can be transmitted over a digital channel.

- There are three steps involved in analog-to-digital encoding, such as:
 - Sampling: In this step, the analog signal is measured at regular intervals, called the sampling rate or sampling frequency. The sampling rate should be at least twice the highest frequency component of the analog signal, according to the Nyquist theorem. The result of sampling is a sequence of samples, each representing the amplitude of the analog signal at a specific point in time.
 - Quantization: In this step, each sample is assigned a discrete value, called a quantization level, from a finite set of possible values. The number of quantization levels depends on the number of bits per sample, which determines the resolution or accuracy of the digitization. The difference between the actual sample value and the quantization level is called the quantization error or noise.
 - Encoding: In this step, each quantization level is encoded into a binary code, using a predefined mapping scheme. The binary code can be transmitted over a digital channel as a digital signal.
- A mnemonic to remember the steps of analog-to-digital encoding is SQE: Sampling, Quantization, Encoding.

Network performance and transmission impairments in Computer Networks

- Network performance is a measure of how well a network can deliver data and services to its users. It can be evaluated by various metrics, such as throughput, delay, jitter, loss, and error rate.
- Throughput is the amount of data successfully transferred over the network in a given time. It depends on the bandwidth, congestion, and protocol efficiency of the network. Higher throughput means better network performance.
- Delay is the time it takes for a packet to travel from the source to the destination. It consists of four components: propagation delay, transmission delay, processing delay, and queuing delay. Lower delay means better network performance.
- Jitter is the variation in delay of packets in a network. It is caused by factors such as network congestion, routing changes, and buffer overflows. High jitter can affect the quality of real-time applications such as voice and video. Lower jitter means better network performance.
- Loss is the percentage of packets that are dropped or lost in the network due to errors, congestion, or timeouts. It can reduce the throughput and reliability of the network. Lower loss means better network performance.

- Error rate is the percentage of packets that are corrupted or altered in the network due to noise, interference, or distortion. It can affect the accuracy and integrity of the data. Lower error rate means better network performance.
- Transmission impairments are any factors that cause the received signal to be different from the transmitted signal. They can degrade the quality and performance of the network. There are three main types of transmission impairments: attenuation, distortion, and noise.
- Attenuation is the loss of signal strength as it travels through the transmission medium. It is caused by factors such as distance, resistance, and interference. Higher attenuation means lower signal quality and higher error rate.
- Distortion is the change in shape or form of the signal as it travels through the transmission medium. It is caused by factors such as frequency-dependent attenuation, phase shift, and nonlinearity. Higher distortion means lower signal quality and higher error rate.
- Noise is the unwanted or random signal that is added to the original signal as it travels through the transmission medium. It is caused by factors such as thermal noise, induced noise, crosstalk noise, and impulse noise. Higher noise means lower signal quality and higher error rate.
- A mnemonic to remember the types of transmission impairments is **Attenuation, Distortion, and Noise, or ADN**.
- A mnemonic to remember the components of delay is **Propagation, Transmission, Processing, and Queuing, or PTPQ**.
- A mnemonic to remember the metrics of network performance is **Throughput, Delay, Jitter, Loss, and Error rate, or TDJLE**.

Switching techniques and multiplexing in Computer Networks

Switching techniques are methods of connecting multiple devices in a network and transferring data between them. Multiplexing is a technique of combining multiple signals into one signal over a shared medium. Switching and multiplexing are essential for efficient and reliable communication in computer networks.

There are three main types of switching techniques: circuit switching, message switching, and packet switching. Each of them has its own advantages and disadvantages, depending on the application and the network requirements.

- **Circuit switching:** In circuit switching, two nodes communicate with each other over a dedicated communication path. The path is established before the data transmission and remains active until the communication is over. The path consists of a series of physical links and switches that connect the source and the destination nodes. Circuit switching is suitable

for real-time and continuous data transmission, such as voice and video calls. However, it has some drawbacks, such as high setup time, low bandwidth utilization, and vulnerability to link failures.

- **Message switching:** In message switching, the whole message is treated as a data unit. The message is stored and forwarded by intermediate nodes until it reaches the destination node. The message may take different paths and arrive out of order. The destination node is responsible for reassembling the message and checking for errors. Message switching is suitable for non-real-time and asynchronous data transmission, such as email and file transfer. However, it has some drawbacks, such as high delay, high storage requirement, and variable delivery time.
- **Packet switching:** The packet switching technique is derived from message switching, where the message is broken down into smaller chunks called packets. Each packet has a header that contains the source and destination addresses, the sequence number, and other information. The packets are routed independently by intermediate nodes based on the network conditions and the routing algorithms. Packet switching is suitable for both real-time and non-real-time data transmission, such as web browsing and streaming. However, it has some drawbacks, such as packet loss, packet delay, and packet reordering.

Multiplexing can be applied to any of the switching techniques to increase the efficiency and capacity of the network. There are different types of multiplexing techniques, such as frequency division multiplexing (FDM), time division multiplexing (TDM), and statistical multiplexing.

- **Frequency division multiplexing (FDM):** In FDM, the available bandwidth of the medium is divided into several frequency bands. Each band is assigned to a different signal. The signals are modulated by different carrier frequencies and combined into one composite signal. The composite signal is transmitted over the medium and demodulated by the receiver. FDM is suitable for analog signals and constant bandwidth applications, such as radio and television broadcasting. However, it has some drawbacks, such as interference, crosstalk, and wastage of bandwidth.
- **Time division multiplexing (TDM):** In TDM, the available time of the medium is divided into several time slots. Each slot is assigned to a different signal. The signals are sampled and multiplexed into one composite signal. The composite signal is transmitted over the medium and demultiplexed by the receiver. TDM is suitable for digital signals and synchronous applications, such as telephone and data networks. However, it has some drawbacks, such as synchronization, overhead, and fixed bandwidth allocation.
- **Statistical multiplexing:** Statistical multiplexing is a communication link sharing technique, which is used in packet switching. The shared

linking is variable in statistical multiplexing, whereas it is fixed in TDM or FDM. This is a strategic application for maximizing the utilization of bandwidth. This can increase the efficiency of the network, as well. Statistical multiplexing is suitable for bursty and asynchronous applications, such as internet and multimedia. However, it has some drawbacks, such as congestion, queuing, and variable delay.

Some mnemonics and learning tricks for the switching techniques and multiplexing are:

- Circuit switching: **C**onstant and **C**ontinuous
- Message switching: **M**essage and **M**emory
- Packet switching: **P**acket and **P**ath
- Frequency division multiplexing: **F**requency and **F**ixed
- Time division multiplexing: **T**ime and **T**imed
- Statistical multiplexing: **S**tatistical and **S**hared

An example of a detailed ASCII diagram for packet switching is:

Source Node	Intermediate Node	Destination Node
+-----+	+-----+	+-----+

Unit 2 - Link layer in Computer Networks and Medium Access Control and Local Area Networks

- The link layer, or layer 2, is the second layer of the OSI model of computer networking. It is responsible for transferring data between nodes on a network segment across the physical layer.
- The link layer is divided into two sublayers: the data link layer and the medium access control (MAC) layer.
- The data link layer provides the functional and procedural means to transfer data between network entities and may also provide error detection and correction. It encapsulates the network layer packets into frames and adds a header and a trailer to each frame.
- The MAC layer is responsible for controlling the access of multiple nodes to a shared medium, such as a wireless channel or a bus network. It coordinates the transmission and reception of frames and avoids or resolves collisions.
- Some of the functions of the link layer are:
 - Framing: dividing the data stream into frames of fixed or variable size.
 - Addressing: adding source and destination addresses to the frames to identify the nodes on the link.
 - Flow control: regulating the rate of data transmission between the sender and the receiver to avoid buffer overflow.
 - Error control: detecting and correcting errors in the frames using techniques such as parity check, checksum, or cyclic redundancy check (CRC).

- Medium access control: determining when and how a node can access the shared medium using protocols such as ALOHA, CSMA/CD, CSMA/CA, or TDMA.
- Link management: establishing, maintaining, and terminating the link between the nodes using protocols such as HDLC, PPP, or SLIP.
- Some of the examples of link layer protocols are:
 - Ethernet: a widely used wired LAN technology that uses CSMA/CD for medium access control and supports various data rates and frame formats.
 - Wi-Fi: a wireless LAN technology that uses CSMA/CA for medium access control and supports various standards such as 802.11a/b/g/n/ac/ax.
 - Bluetooth: a wireless personal area network (PAN) technology that uses frequency hopping spread spectrum (FHSS) for medium access control and supports various profiles such as headset, keyboard, or mouse.
 - ATM: a high-speed switched network technology that uses fixed-size cells of 53 bytes and supports various quality of service (QoS) classes and traffic types.
- Some of the mnemonics and learning tricks for the link layer are:
 - To remember the functions of the link layer, use the acronym FAMEL: Framing, Addressing, Medium access control, Error control, Link management.
 - To remember the types of errors that can occur in the link layer, use the acronym BUD: Bit errors, Undetected errors, Dropped frames.
 - To remember the types of medium access control protocols, use the acronym CAT: Contention-based, Allocation-based, Token-based.

Link layer in Computer Networks

- The link layer is the lowest layer in the Internet protocol suite, the networking architecture of the Internet.
- The link layer is the group of methods and communications protocols confined to the link that a host is physically connected to.
- The link layer is responsible for transferring data between nodes on a network segment across the physical layer.
- The link layer may also provide the means to detect and possibly correct errors that can occur in the physical layer.
- The link layer is concerned with local delivery of frames between nodes on the same level of the network.
- The link layer usually provides and gives data reliability and provides various tools to establish, maintain, and also release data link connections between network nodes.
- The link layer can be divided into two sublayers: the logical link control (LLC) sublayer and the media access control (MAC) sublayer.
- The LLC sublayer provides services such as flow control, error control, and

multiplexing to the upper layers.

- The MAC sublayer provides services such as addressing, channel access, and framing to the LLC sublayer.
- The link layer can use different protocols depending on the type of link, such as Ethernet, Wi-Fi, Bluetooth, PPP, etc.
- The link layer can also use different techniques to handle issues such as collision, congestion, and synchronization, such as CSMA/CD, CSMA/CA, token ring, etc.

Some mnemonics and learning tricks for the link layer are:

- To remember the functions of the link layer, use the acronym **FAE** (Framing, Addressing, Error control).
- To remember the sublayers of the link layer, use the acronym **LM** (Logical link control, Media access control).
- To remember the types of link layer protocols, use the acronym **EWBPP** (Ethernet, Wi-Fi, Bluetooth, PPP).
- To remember the types of link layer techniques, use the acronym **CTC** (Collision, Congestion, Synchronization).

Framing in link layer in Computer Networks

- Framing is a function of the data link layer. It provides a way for a sender to transmit a set of bits that are meaningful to the receiver .
- Framing uses frames to send or receive data. The data link layer receives packets from the network layer and converts them into frames .
- Frames have headers that contain information such as error-checking codes, source and destination addresses, and control information .
- Frames are the result of the final layer of encapsulation before the data is transmitted over the physical layer.
- Framing is necessary because the physical layer only accepts and transfers a stream of bits without any regard to meaning or structure.
- Framing also helps to synchronize the transmission and reception of data, and to detect and correct errors .
- There are different types of framing methods, such as:
 - Character count: The header of the frame contains a field that specifies the number of characters in the frame. This method is simple but unreliable, as any error in the count field or the data can cause framing errors .
 - Byte stuffing: The frame starts and ends with a special byte, such as DLE STX (Data Link Escape, Start of Text) and DLE ETX (Data Link Escape, End of Text). If the data contains the same byte as

the delimiter, it is preceded by another DLE byte to escape it. This method is more reliable but requires extra bytes for stuffing .

- Bit stuffing: The frame starts and ends with a special bit pattern, such as 01111110. If the data contains five consecutive 1s, a 0 is inserted after them to avoid confusion with the delimiter. This method is also reliable but requires extra bits for stuffing .
 - Physical layer coding violations: The frame uses the physical layer encoding scheme, such as Manchester or NRZ, to mark the start and end of the frame. For example, a violation of the Manchester code, such as 00 or 11, can indicate the boundary of the frame. This method does not require extra bits or bytes, but depends on the physical layer encoding .
- A possible mnemonic to remember the types of framing methods is: **Charlie Bought Bit Pies** (Character count, Byte stuffing, Bit stuffing, Physical layer coding violations).
 - A possible learning trick to understand the concept of framing is to imagine a conveyor belt that carries boxes of different sizes and shapes. The boxes represent the frames, and the conveyor belt represents the physical layer. The boxes have labels that indicate the contents, the sender, and the receiver of the box. The labels also have a barcode that can be scanned to check for errors. The boxes have different ways of indicating where they start and end, such as using a specific color, a specific shape, a specific pattern, or a specific mark. These ways represent the different framing methods. The conveyor belt moves the boxes from one end to the other, without caring about what is inside the boxes or how they are marked. The receiver has to identify the boxes that are meant for it, and open them to get the contents. The receiver also has to scan the barcode to verify that the box is not damaged or corrupted. This is how framing works in the data link layer.

Error Detection and Correction in link layer in Computer Networks

- Error detection and correction in link layer is the process of detecting and correcting data frames that have been corrupted or lost during transmission .
- Error detection and correction is important to ensure reliable and accurate data communication in computer networks .
- The basic approach for error detection and correction is the use of redundancy, where additional bits are added to the data to facilitate the detection and correction of errors .
- There are three main techniques for error detection in link layer: parity check, checksum and cyclic redundancy check (CRC) .
- Parity check is the simplest technique, where an extra bit, called parity bit, is added to the data to make the number of 1s either even (in case of

even parity) or odd (in case of odd parity) .

- Parity check can detect single-bit errors, but not multiple-bit errors.
- Checksum is a technique where the sender computes a value, called checksum, by adding the data bytes and sends it along with the data .
- The receiver computes the checksum of the received data and compares it with the received checksum .
- If the checksums match, the data is assumed to be error-free, otherwise an error is detected .
- Checksum can detect multiple-bit errors, but not all of them .
- CRC is a technique where the sender and the receiver agree on a polynomial, called generator polynomial, and use it to generate a value, called CRC, by dividing the data by the polynomial .
- The sender appends the CRC to the data and sends it to the receiver .
- The receiver divides the received data by the same polynomial and checks if the remainder is zero .
- If the remainder is zero, the data is assumed to be error-free, otherwise an error is detected .
- CRC can detect multiple-bit errors and burst errors (where a sequence of bits are corrupted) .
- Error correction is the process of identifying and correcting the corrupted bits in the data .
- There are two main techniques for error correction in link layer: automatic repeat request (ARQ) and forward error correction (FEC) .
- ARQ is a technique where the sender and the receiver use acknowledgements (ACKs) and negative acknowledgements (NAKs) to communicate the status of the received data .
- The sender sends a data frame and waits for an ACK from the receiver .
- The receiver sends an ACK if the data frame is error-free, or a NAK if an error is detected .
- The sender retransmits the data frame if it does not receive an ACK or receives a NAK within a specified time .
- ARQ can correct errors by retransmission, but it requires feedback and increases the delay .
- FEC is a technique where the sender adds extra bits, called error-correcting codes, to the data to enable the receiver to correct errors without retransmission .
- The receiver uses the error-correcting codes to detect and correct errors in the data .
- FEC can correct errors without feedback and reduce the delay, but it requires more bandwidth and computation .
- There are different types of error-correcting codes, such as Hamming code, Reed-Solomon code, and convolutional code .
- Hamming code is a technique where the sender adds extra bits, called parity bits, to the data to form a code word that has a certain Hamming distance .
- The Hamming distance is the number of bits that differ between two code

words .

- The receiver uses the parity bits to detect and correct errors in the data by finding the closest code word [

Flow control in link layer in Computer Networks

- Flow control is a technique that allows two stations working at different speeds to communicate with each other.
- It regulates the amount of data that a sender can send so that a fast sender does not overwhelm a slow receiver .
- It makes the sender wait until an acknowledgment is received from the receiver's end .
- Methods of flow control are stop-and-wait, and sliding window .

Stop-and-wait

- In this method, the sender sends one frame at a time and waits for an acknowledgment from the receiver before sending the next frame.
- The receiver sends an acknowledgment after receiving and processing the frame.
- The sender can send the next frame only after receiving the acknowledgment.
- This method is simple but inefficient, as the sender has to wait for a round-trip time (RTT) between each frame.
- The utilization of the channel is low, as the channel is idle during the waiting time.

Sliding window

- In this method, the sender can send multiple frames without waiting for acknowledgments, as long as the number of frames does not exceed a predefined window size.
- The receiver sends an acknowledgment for each frame, indicating the next expected frame.
- The sender slides the window forward as it receives acknowledgments, and can send new frames that fall within the window.
- This method is more efficient and utilizes the channel better, as the sender does not have to wait for a RTT between each frame.
- The window size can be adjusted dynamically based on the network conditions and the receiver's buffer capacity.

Mnemonics and learning tricks

- To remember the difference between stop-and-wait and sliding window, you can use the following mnemonics:
 - Stop-and-wait: SAW (Slow And Wasteful)
 - Sliding window: SW (Swift and Wise)

- To remember the formula for the utilization of the channel in stop-and-wait, you can use the following trick:
 - Utilization = $1 / (1 + 2a)$, where $a = \text{RTT} / \text{transmission time}$
 - Think of a as the ratio of the waiting time to the sending time
 - The higher the a , the lower the utilization
 - The lower the a , the higher the utilization
 - When $a = 0$, the utilization is 1 (maximum)
 - When $a = \text{infinity}$, the utilization is 0 (minimum)

Elementary Data Link Protocols in link layer in Computer Networks

- Protocols in the data link layer are designed to perform the basic functions of framing, error control and flow control.
- Framing is the process of dividing bit-streams from the physical layer into data frames whose size ranges from a few hundred to a few thousand bytes.
- Error control is the process of detecting and correcting errors that may occur during transmission or reception of data frames.
- Flow control is the process of regulating the rate of data transmission between the sender and the receiver to avoid congestion or buffer overflow.
- Elementary data link layer protocols are the simplest protocols that can be used for data transmission over a single link.
- Elementary data link layer protocols are divided into three categories, as given below:
 - Protocol 1: Unrestricted simplex protocol
 - Protocol 2: Simplex stop and wait protocol
 - Protocol 3: Simplex protocol for noisy channels.
- Let us discuss each protocol one by one.

Protocol 1: Unrestricted simplex protocol

- This protocol is used for one-way data transmission over a noiseless channel.
- The sender continuously sends data frames without waiting for any acknowledgment from the receiver.
- The receiver simply accepts and processes the incoming data frames.
- There is no error or flow control in this protocol.
- This protocol is suitable for applications where data loss is not critical, such as streaming audio or video.
- A possible mnemonic for this protocol is: **U**nrestricted **S**implex **P**rotocol = **U**nlimited **S**ending **P**ermission.

Protocol 2: Simplex stop and wait protocol

- This protocol is used for one-way data transmission over a noisy channel.
- The sender sends one data frame at a time and waits for an acknowledgment from the receiver before sending the next frame.
- The receiver sends an acknowledgment after receiving and processing a data frame.
- If the sender does not receive an acknowledgment within a specified time, it assumes that the frame or the acknowledgment was lost and retransmits the frame.
- This protocol provides error control but not flow control.
- This protocol is suitable for applications where data loss is not acceptable, such as file transfer or email.
- A possible mnemonic for this protocol is: **S**implex **S**top and **W**ait **P**rotocol = **S**end **S**ingle **W**ait **P**atiently.

Protocol 3: Simplex protocol for noisy channels

- This protocol is an improvement over protocol 2 that can handle the problem of duplicate frames.
- The sender and the receiver use sequence numbers to identify each data frame and acknowledgment.
- The sender sends one data frame with a sequence number and waits for an acknowledgment with the same sequence number before sending the next frame.
- The receiver sends an acknowledgment with the sequence number of the last correctly received frame.
- If the sender receives an acknowledgment with a different sequence number, it assumes that the frame or the acknowledgment was lost or duplicated and retransmits the frame with the same sequence number.
- This protocol provides error control but not flow control.
- This protocol is suitable for applications where data loss and duplication are not acceptable, such as database transactions or online banking.
- A possible mnemonic for this protocol is: **S**implex **P**rotocol for **N**oisy **C**hannels = **S**end **P**roper **N**umber **C**heck.

Sliding Window protocols in link layer in Computer Networks

- The sliding window protocol is a data link layer protocol that is useful in the sequential and reliable delivery of the data frames .
- Using the sliding window protocol, the sender can send multiple frames at a time before receiving an acknowledgment from the receiver .
- The sliding window protocol uses a mechanism of sequence numbers to identify and order the frames .
- The sender and the receiver maintain a window of frames that can be sent or received at a time .

- The window size is determined by the available buffer space and the bandwidth-delay product.
- The window slides forward when the sender receives an acknowledgment or the receiver sends an acknowledgment .
- There are two types of sliding window protocols: stop-and-wait and go-back-N .
- In stop-and-wait protocol, the sender sends one frame at a time and waits for an acknowledgment before sending the next frame .
- In go-back-N protocol, the sender can send up to N frames at a time without waiting for an acknowledgment, where N is the window size .
- The sliding window protocol is also used in Transmission Control Protocol (TCP), which operates at the transport layer .

Advantages of sliding window protocol

- It improves the efficiency and throughput of data transmission by allowing multiple frames to be sent at a time .
- It provides flow control and error control by using acknowledgments and sequence numbers .
- It adapts to the varying network conditions by adjusting the window size dynamically.

Disadvantages of sliding window protocol

- It may cause duplication of frames due to lost or delayed acknowledgments .
- It may cause wastage of bandwidth and resources due to retransmission of frames in case of errors .
- It may cause deadlock or starvation if the sender or the receiver does not send or receive any frames for a long time .

Examples of sliding window protocol

- The following is an example of stop-and-wait protocol, where the sender sends one frame at a time and waits for an acknowledgment before sending the next frame. The window size is 1 for both the sender and the receiver.

Sender	Receiver
S0	
	R0
S1	
	R1
S2	
	R2
S3	
	R3

- The following is an example of go-back-N protocol, where the sender can send up to N frames at a time without waiting for an acknowledgment, where N is the window size. The window size is 3 for the sender and 1 for the receiver. The sender retransmits all the frames from the last unacknowledged frame in case of an error.

Sender	Receiver
S0	
S1	
S2	
	R0
S3	
	R1
S4	
	R2
S5	
	R3
S6	
	R4
S7	
	R5
S8	
	R6
	Error
S6	
S7	
S8	
	R6
	R7
	R8

Mnemonics and learning tricks for sliding window protocol

Medium Access Control and Local Area Networks

- Medium Access Control (MAC) is a sublayer of the data link layer that defines how multiple devices share the same medium (such as a cable or a wireless channel) to communicate in a local area network (LAN).
- MAC protocols are responsible for addressing, framing, error control, flow control, and channel access in LANs.
- There are two main approaches to MAC protocols in LANs: contention-based and token-passing.
 - Contention-based protocols allow any device to transmit whenever

the channel is idle, but they have to deal with collisions (when two or more devices transmit at the same time) and retransmissions. Examples of contention-based protocols are Carrier Sense Multiple Access with Collision Detection (CSMA/CD), used in Ethernet, and Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA), used in wireless LANs.

- Token-passing protocols use a special frame called a token that circulates among the devices in a logical ring or bus topology. Only the device that holds the token can transmit, and it has to pass the token to the next device after transmission. This avoids collisions, but introduces some delay and overhead. Examples of token-passing protocols are Token Ring and Fiber Distributed Data Interface (FDDI).
- MAC addresses are unique identifiers assigned to each device in a LAN. They are usually 48 bits long and written in hexadecimal notation, such as 00:0A:95:9D:68:16. MAC addresses are used to deliver frames to the correct destination in a LAN, and they are also used for filtering and security purposes.
- MAC protocols can also support quality of service (QoS) requirements for different types of applications, such as voice, video, and data. QoS refers to the ability of a network to provide different levels of performance guarantees, such as bandwidth, delay, jitter, and packet loss. Some MAC protocols that support QoS are IEEE 802.11e for wireless LANs, IEEE 802.1Q for VLANs, and IEEE 802.1p for priority tagging.

Some mnemonics and learning tricks for MAC and LANs are:

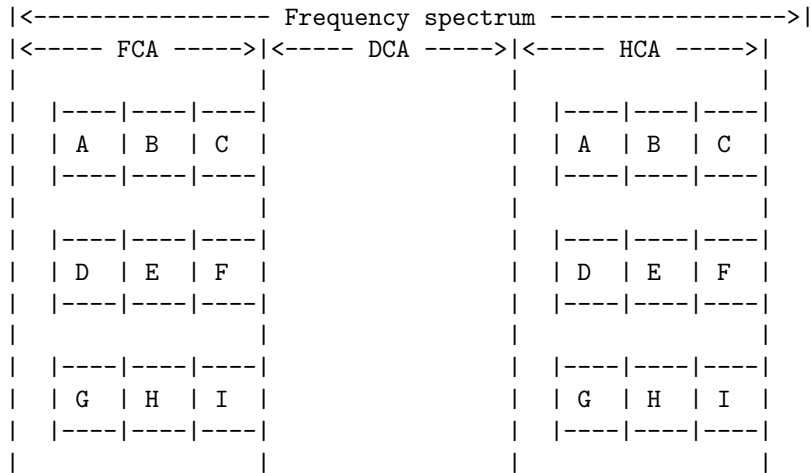
- MAC is the acronym for Medium Access Control, and it also sounds like “make”, which reminds us that MAC protocols make the communication possible in a LAN.
- CSMA/CD and CSMA/CA are the two most common contention-based protocols, and they can be remembered by the words “collision detection” and “collision avoidance”. CD means that the devices detect collisions and back off for a random time before retransmitting. CA means that the devices avoid collisions by using a handshake mechanism before transmitting.
- Token-passing protocols use a token to control the access to the channel, and they can be remembered by the word “token”. A token is a small piece of data that represents the right to transmit, and it is passed from one device to another in a logical ring or bus.
- MAC addresses are unique identifiers for each device in a LAN, and they can be remembered by the word “address”. An address is a way of locating someone or something, and MAC addresses are used to locate the destination device in a LAN.

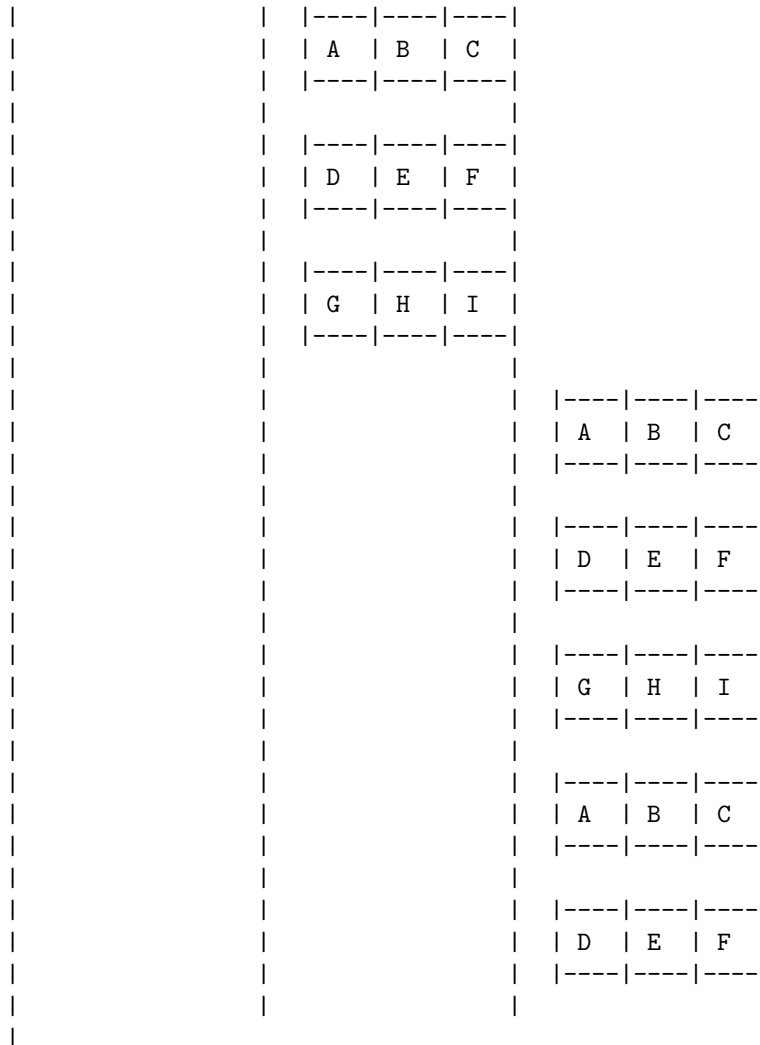
Channel allocation in medium access control

- Channel allocation is the process of assigning a portion of the available

bandwidth to different users or devices in a communication network.

- Channel allocation can be classified into three types: fixed, dynamic, and hybrid.
- Fixed channel allocation (FCA) assigns a fixed number of channels to each user or device, regardless of their traffic demand or activity. FCA is simple and efficient, but it may result in underutilization or overutilization of the channels.
- Dynamic channel allocation (DCA) assigns channels to users or devices on demand, based on their traffic demand or activity. DCA is flexible and adaptive, but it may incur higher overhead and complexity.
- Hybrid channel allocation (HCA) combines the advantages of FCA and DCA by using both fixed and dynamic channels. HCA can achieve better performance and efficiency, but it may require more coordination and synchronization.
- A mnemonic to remember the types of channel allocation is: **Fix Dynamic Hybrid** (FDH).
- An example of FCA is frequency division multiple access (FDMA), which divides the frequency spectrum into fixed channels and assigns them to different users or devices.
- An example of DCA is carrier sense multiple access (CSMA), which allows users or devices to sense the channel before transmitting and avoid collisions with other users or devices.
- An example of HCA is time division multiple access (TDMA), which divides the time into fixed slots and assigns them to different users or devices, but also allows dynamic allocation of slots within a frame.
- A diagram to illustrate the types of channel allocation is:





Multiple access protocols in medium access control

- Multiple access protocols are a set of protocols operating in the Medium Access Control sublayer (MAC sublayer) of the Open Systems Interconnection (OSI) model .
- These protocols allow a number of nodes or users to access a shared network channel, such as a wireless channel or a bus network, without interfering with each other or causing collisions .
- The main design criteria for multiple access protocols are efficiency, fairness, reliability, and scalability.

- Multiple access protocols can be classified into three categories: random access, controlled access, and channelization .
 - Random access protocols: In these protocols, all stations have equal priority and can send data depending on the medium's state (idle or busy). There is no fixed time for sending data and no central controller to coordinate the transmissions. Examples of random access protocols are ALOHA, Carrier Sense Multiple Access (CSMA), CSMA with Collision Avoidance (CSMA/CA), and CSMA with Collision Detection (CSMA/CD) .
 - Controlled access protocols: In these protocols, the stations have to follow some rules or obtain permission before sending data. There is a central controller or a distributed algorithm that decides which station can access the channel at a given time. Examples of controlled access protocols are Reservation, Polling, and Token Passing .
 - Channelization protocols: In these protocols, the channel is divided into smaller sub-channels that can be allocated to different stations. The sub-channels can be based on time, frequency, code, or space. Examples of channelization protocols are Time Division Multiple Access (TDMA), Frequency Division Multiple Access (FDMA), Code Division Multiple Access (CDMA), and Space Division Multiple Access (SDMA) .
- Some mnemonics and learning tricks for the multiple access protocols are:
 - ALOHA: ALOHA is a Hawaiian word that means hello and goodbye. It is also an acronym for Additive Links On-line Hawaii Area. ALOHA is a simple random access protocol that does not sense the channel before sending data. It relies on acknowledgments and timeouts to detect and resolve collisions .
 - CSMA: CSMA stands for Carrier Sense Multiple Access. It is a random access protocol that senses the channel before sending data. If the channel is busy, it waits for a random time and tries again. It can be further classified into three types: 1-persistent, non-persistent, and p-persistent .
 - CSMA/CA: CSMA/CA stands for Carrier Sense Multiple Access with Collision Avoidance. It is a random access protocol that tries to avoid collisions by using a handshake mechanism. Before sending data, it sends a Request to Send (RTS) frame and waits for a Clear to Send (CTS) frame from the receiver. It is used in wireless networks such as IEEE 802.11 (Wi-Fi) .
 - CSMA/CD: CSMA/CD stands for Carrier Sense Multiple Access with Collision Detection. It is a random access protocol that tries to detect collisions by monitoring the channel while sending data. If a collision is detected, it stops the transmission and waits for a random time before trying again. It is used in wired networks such as Ethernet .

- Reservation: Reservation is a controlled access protocol that uses a reservation frame to reserve the channel for a station. The reservation frame contains a bitmap that indicates which station wants to send data in the next frame. It is used in wireless networks such as IEEE 802.16 (WiMAX) .
- Polling: Polling is a controlled access protocol that uses a central controller or a master station to poll the other stations or slaves for their data requests. The master station grants the channel to one slave at a time and waits for its data. It is used in networks such as Bluetooth and USB .
- Token Passing: Token Passing is a controlled access protocol that uses a special frame called a token to grant the channel to a station. The token circulates among the stations in a logical ring or a bus. The station that has the token can send data and then passes the token to

LAN standards in local area network

- LAN standards are the specifications and protocols that define how devices communicate and share data over a local area network (LAN).
- LAN standards are developed and maintained by various organizations, such as the Institute of Electrical and Electronics Engineers (IEEE), the International Organization for Standardization (ISO), and the Internet Engineering Task Force (IETF).
- LAN standards cover different aspects of LAN communication, such as physical layer, data link layer, network layer, and application layer.
- Some of the most common LAN standards are:
 - Ethernet: The most widely used LAN standard, which defines how devices transmit and receive data over a shared medium, such as twisted pair, coaxial cable, or optical fiber. Ethernet uses the Carrier Sense Multiple Access with Collision Detection (CSMA/CD) protocol to avoid data collisions and ensure reliable data transfer. Ethernet supports various data rates, such as 10 Mbps, 100 Mbps, 1 Gbps, and 10 Gbps. Ethernet is standardized by the IEEE 802.3 series of standards.
 - Wi-Fi: The most popular wireless LAN standard, which defines how devices communicate over radio waves using the IEEE 802.11 series of standards. Wi-Fi supports various data rates, such as 11 Mbps, 54 Mbps, 600 Mbps, and 1.3 Gbps. Wi-Fi uses the Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA) protocol to avoid data collisions and ensure reliable data transfer. Wi-Fi also provides security features, such as encryption, authentication, and access control, using the IEEE 802.11i standard.

- Token Ring: A LAN standard that uses a token-passing protocol to regulate data transmission over a ring topology. Token Ring supports data rates of 4 Mbps and 16 Mbps. Token Ring is standardized by the IEEE 802.5 standard.
 - FDDI: A LAN standard that uses a token-passing protocol to regulate data transmission over a dual ring topology. FDDI supports data rates of 100 Mbps and uses optical fiber as the transmission medium. FDDI is standardized by the ANSI X3T9.5 standard.
 - ATM: A LAN standard that uses fixed-length cells to transmit data over various types of media, such as twisted pair, coaxial cable, optical fiber, or wireless. ATM supports data rates of up to 622 Mbps and provides quality of service (QoS) features, such as guaranteed bandwidth, delay, and jitter. ATM is standardized by the ITU-T I.150 series of standards.
- A mnemonic to remember some of the LAN standards is: **Every Wizard Takes Fancy Apples**. (Ethernet, Wi-Fi, Token Ring, FDDI, ATM)

Link layer switches & bridges in local area network

- Link layer switches and bridges are network devices that operate at the data link layer (layer 2) of the OSI model.
- They connect multiple LANs (local area networks) together to form a larger LAN or a single broadcast domain.
- They use MAC addresses to forward Ethernet frames from one device to another device in an Ethernet standard based LAN.
- They store and forward frames, which means they receive the entire frame, check for errors, and then forward it to the appropriate destination or drop it if it is not valid.
- They can also filter frames based on MAC addresses, which reduces the network congestion and improves the performance.
- They can learn MAC addresses dynamically by observing the source addresses of the incoming frames and updating their MAC address table accordingly.
- They can also handle different data link layer technologies, such as Ethernet, FDDI, Token Ring, etc. by converting the frame format and adjusting the maximum frame size.
- They can be classified into two types: transparent bridges and source routing bridges.
 - Transparent bridges are the most common type of bridges. They use the destination MAC address of the frame to determine the outgoing port. They use the spanning tree protocol (STP) to avoid loops in the network topology.
 - Source routing bridges are used in Token Ring networks. They use the source MAC address of the frame and a routing information field (RIF) to determine the path to the destination. They do not use

- STP, but rely on the source device to avoid loops.
- Link layer switches are similar to bridges, but they have some advantages over them, such as:
 - They have multiple ports and can forward frames in parallel, which increases the bandwidth and reduces the latency.
 - They have larger MAC address tables and can handle more devices and traffic.
 - They can perform more advanced functions, such as VLANs, QoS, security, etc. by using additional information in the frame header or payload.

Learning bridge algorithms in local area network

- A bridge is a device that connects two or more local area networks (LANs) at the data link layer and forwards frames between them based on the destination address.
- A bridge algorithm is a set of rules that determines how a bridge learns the addresses of the devices connected to the LANs and how it decides which frames to forward or discard.
- There are two main types of bridge algorithms: spanning tree and source routing.
- Spanning tree algorithm :
 - It creates a loop-free logical topology of the network by disabling some of the bridge ports.
 - It elects a root bridge and assigns a cost to each bridge port based on the bandwidth of the link.
 - It uses the Bridge Protocol Data Units (BPDUs) to exchange information among bridges and to detect topology changes.
 - It forwards frames based on the destination address and the port cost.
 - It has the advantages of simplicity, transparency, and adaptability to topology changes.
 - It has the disadvantages of slow convergence, suboptimal routing, and limited scalability.
- Source routing algorithm :
 - It allows the source device to specify the path of the frame through the network by adding a routing information field (RIF) to the frame header.
 - It uses the All Routes Explorer (ARE) and the Spanning Tree Explorer (STE) frames to discover and update the routes in the network.
 - It forwards frames based on the RIF and the destination address.
 - It has the advantages of fast convergence, optimal routing, and high scalability.
 - It has the disadvantages of complexity, overhead, and security risks.
- A mnemonic to remember the difference between spanning tree and source routing is: **ST** stands for **S**imple and **T**ransparent, while **SR** stands for

Specific and Rapid.

Spanning Tree Algorithms in Local Area Network

- Spanning tree algorithms are used to prevent loops in a network topology that has redundant links between switches or bridges.
- Loops can cause broadcast storms, multiple frame copies, and MAC address table instability.
- Spanning tree algorithms create a logical tree structure from the physical mesh topology by blocking some links and enabling others.
- The root of the tree is a designated switch or bridge called the root bridge, which is elected based on a priority value and a MAC address.
- The links that are not blocked are called the spanning tree, and they provide a loop-free path between any two nodes in the network.
- There are different versions of spanning tree protocols, such as STP, RSTP, MSTP, and PVSTP, that have different features and enhancements.
- Some of the advantages of spanning tree algorithms are:
 - They provide network redundancy and fault tolerance by activating backup links in case of a failure.
 - They simplify the network design and management by eliminating the need for manual configuration of loop prevention.
 - They improve network performance and efficiency by reducing broadcast traffic and preventing congestion.
- Some of the disadvantages of spanning tree algorithms are:
 - They introduce convergence delays when the network topology changes, which can affect time-sensitive applications and protocols.
 - They waste bandwidth and resources by blocking some links that could be used for load balancing or aggregation.
 - They can cause suboptimal routing and increased hop count by forcing traffic to follow the tree structure instead of the shortest path.
- Some of the mnemonics and learning tricks for spanning tree algorithms are:
 - To remember the order of criteria for selecting the root bridge, use the acronym B.A.D: Bridge priority, Address (MAC), Default priority.
 - To remember the order of criteria for selecting the root port on a non-root bridge, use the acronym B.A.D.C: Bridge ID of the root bridge, Aggregate cost to the root bridge, Designated bridge ID, Designated port ID.
 - To remember the order of criteria for selecting the designated port on a segment, use the acronym B.A.P: Bridge ID of the advertising bridge, Aggregate cost to the root bridge, Port ID of the advertising port.

Unit 3 - Network Layer in Computer Networks

- The network layer is the third layer in the OSI model and the internet layer in the TCP/IP model.
- The network layer is responsible for transmitting data segments between networks in the form of packets.
- The network layer assigns source and destination IP addresses to the data segments and determines the best paths for data delivery using routing protocols.
- The network layer also manages sub-networks and internetworking, which are the processes of connecting different types of networks using routers, gateways, or bridges.
- The network layer provides logical addressing, which is independent of the physical addressing used by the data link layer.
- The network layer can also perform fragmentation, which is the division of large packets into smaller ones to fit the maximum transmission unit (MTU) of the underlying network.
- The network layer can also perform reassembly, which is the opposite of fragmentation, and involves putting the fragmented packets back together at the destination.
- The network layer can also perform error control, which is the detection and correction of errors that may occur during transmission.
- The network layer can also perform congestion control, which is the prevention or reduction of network congestion by regulating the flow of packets.
- The network layer can also perform quality of service (QoS), which is the provision of different levels of service to different types of traffic based on their requirements and priorities.

Some examples of network layer protocols are:

- Internet Protocol (IP): The main protocol of the network layer that provides logical addressing and routing for packets.
- Internet Control Message Protocol (ICMP): A protocol that sends and receives error and control messages between hosts and routers.
- Internet Group Management Protocol (IGMP): A protocol that manages multicast groups of hosts and routers.
- Address Resolution Protocol (ARP): A protocol that maps IP addresses to MAC addresses of the data link layer.
- Reverse Address Resolution Protocol (RARP): A protocol that maps MAC addresses to IP addresses of the network layer.
- Internet Protocol version 6 (IPv6): The latest version of IP that provides a larger address space and improved features.

Some mnemonics and learning tricks for the network layer are:

- To remember the main functions of the network layer, use the acronym **PARFERQ** (Pronounced as “par-fair-q”), which stands for **P**acketization, **A**ddressing, **R**outing, **F**ragmentation, **E**rror control,

Reassembly, and QoS.

- To remember the difference between logical and physical addressing, use the analogy of a **house**. The logical address is like the **postal address** of the house, which is unique and independent of the location. The physical address is like the **GPS coordinates** of the house, which depend on the location and may change if the house is moved.
- To remember the difference between fragmentation and reassembly, use the analogy of a **pizza**. Fragmentation is like **cutting** a large pizza into smaller slices to fit into a box. Reassembly is like **putting** the slices back together to form the original pizza.

Point-to-point networks in network layer

- A point-to-point network is a network topology that connects two nodes directly using a single link.
- A point-to-point network can be used to connect two routers, two hosts, or a router and a host.
- A point-to-point network can be implemented using different physical media, such as copper wire, fiber optic cable, radio waves, or satellite links.
- A point-to-point network can use different data link layer protocols, such as High-level Data Link Control (HDLC), Point-to-Point Protocol (PPP), or Point-to-Point Tunneling Protocol (PPTP).
- A point-to-point network can support different network layer protocols, such as Internet Protocol (IP), Internet Protocol version 6 (IPv6), or Internetwork Packet Exchange (IPX).
- A point-to-point network can provide various features, such as loop connection authentication, transmission encryption, data compression, error detection, and error correction.

Advantages of point-to-point networks in network layer

- A point-to-point network is simple and easy to configure, as it does not require any intermediate devices or complex routing algorithms.
- A point-to-point network is reliable and efficient, as it does not suffer from congestion, collisions, or interference from other nodes.
- A point-to-point network is secure and private, as it can use encryption and authentication mechanisms to protect the data from unauthorized access or modification.
- A point-to-point network is flexible and scalable, as it can support different types of data and protocols, and can be extended by adding more links and nodes.

Disadvantages of point-to-point networks in network layer

- A point-to-point network is costly and wasteful, as it requires a dedicated link for each pair of nodes, which may not be fully utilized or available at all times.

- A point-to-point network is limited and isolated, as it does not allow communication with other nodes or networks, unless a gateway or a bridge is used.
- A point-to-point network is vulnerable and fragile, as it depends on the availability and quality of the single link, which may be damaged, disrupted, or compromised.

Examples of point-to-point networks in network layer

- A dial-up connection between a computer and an Internet service provider (ISP) using a modem and a telephone line.
- A leased line connection between two offices using a dedicated fiber optic cable.
- A wireless connection between two laptops using a Bluetooth or Wi-Fi adapter.
- A satellite connection between a remote site and a central hub using a dish antenna and a transceiver.

ASCII diagram of a point-to-point network in network layer

```

+-----+      +-----+
| Node |-----| Node |
+-----+      +-----+

```

Logical addressing in network layer

- Logical addressing is a method of identifying hosts and routers in a network using addresses that are independent of the physical location and connection of the devices.
- Logical addresses are assigned by the network layer protocol, such as IP (Internet Protocol), and are used to route packets from the source to the destination across multiple networks.
- Logical addresses are also called network addresses or layer 3 addresses, as they belong to the network layer of the OSI model.
- Logical addresses have two components: a network ID and a host ID. The network ID identifies the network to which the device belongs, and the host ID identifies the device within that network.
- Logical addresses are hierarchical, meaning that they can be divided into subnets of different sizes, depending on the network requirements. Subnetting allows more efficient use of the address space and reduces the size of the routing tables.
- Logical addresses are usually represented in a human-readable format, such as dotted decimal notation for IPv4 (e.g., 192.168.1.1) or hexadecimal notation for IPv6 (e.g., 2001:db8::1).
- Logical addresses are mapped to physical addresses, such as MAC addresses, using address resolution protocols, such as ARP (Address Resolution Protocol) for IPv4 or NDP (Neighbor Discovery Protocol) for IPv6.

This allows the data link layer to deliver the packets to the correct device on the same network.

- Logical addressing is essential for the functioning of the internet, as it enables communication between devices that are not directly connected or located in different networks. Logical addressing also provides flexibility and scalability for network growth and changes.

Basic internetworking in network layer

- Internetworking is the process of connecting different networks using routers and other devices to form a larger network that can exchange data across different protocols and technologies.
- The network layer is responsible for providing logical addressing, routing, and packet forwarding between different networks.
- The standard reference model for internetworking is the Open Systems Interconnection (OSI) model, which defines seven layers of communication functions. The network layer is the third layer in this model.
- The most notable example of internetworking is the Internet, which is a global network of networks that uses the Internet Protocol (IP) as the common network layer protocol.
- There are three main types of internetworking: Internet, Intranet, and Extranet.
 - Internet: A public network that connects millions of networks and devices worldwide using IP and other protocols. Anyone can access the Internet through an Internet Service Provider (ISP) or a public network access point.
 - Intranet: A private network that connects networks and devices within an organization using IP and other protocols. Only authorized users can access the intranet through a firewall or a virtual private network (VPN).
 - Extranet: A private network that connects networks and devices of different organizations using IP and other protocols. Only authorized users can access the extranet through a firewall or a VPN. An extranet can be used for business-to-business (B2B) communication, collaboration, or e-commerce.
- There are three kinds of internetwork addresses that are commonly used: data-link layer addresses, Media Access Control (MAC) addresses, and network-layer addresses.
 - Data-link layer addresses: These are physical addresses that uniquely identify each network interface of a device. They are usually assigned by the manufacturer of the device and are stored in the hardware. Data-link layer addresses are used for communication within a single network.
 - MAC addresses: These are a type of data-link layer addresses that are 48 bits long and are written in hexadecimal format. MAC addresses are used for communication within Ethernet, Wi-Fi, and other net-

works that use the IEEE 802 standards.

- Network-layer addresses: These are logical addresses that uniquely identify each device or network on an internetwork. They are usually assigned by the network administrator or the ISP and are stored in the software. Network-layer addresses are used for communication across different networks. IP addresses are a type of network-layer addresses that are 32 bits long (IPv4) or 128 bits long (IPv6) and are written in decimal or hexadecimal format.
- A mnemonic to remember the difference between data-link layer addresses and network-layer addresses is: **D**ata-link layer addresses are **D**evice-specific, while **N**etwork-layer addresses are **N**etwork-specific.
- A mnemonic to remember the difference between MAC addresses and IP addresses is: **M**AC addresses are **M**anufacturer-assigned, while **I**P addresses are **I**nternet-assigned.

IP

- IP stands for Internet Protocol, which is a set of rules that governs how data packets are transmitted over a network.
- IP is one of the core protocols of the TCP/IP suite, which is the most widely used network protocol stack in the world.
- IP is responsible for addressing, routing, and fragmenting data packets across different networks and devices.
- IP operates at the network layer (layer 3) of the OSI model, which means it can work with any type of link layer (layer 2) protocol, such as Ethernet, Wi-Fi, or PPP.
- IP has two main versions: IPv4 and IPv6, which differ in the size and format of the addresses they use.
- IPv4 uses 32-bit addresses, which can support up to 4.3 billion unique devices on the network. However, due to the rapid growth of the Internet, IPv4 addresses are running out, and many devices have to share the same address using techniques like NAT (Network Address Translation).
- IPv6 uses 128-bit addresses, which can support up to 3.4×10^{38} unique devices on the network. IPv6 also has some other features and improvements over IPv4, such as simplified header format, enhanced security, and support for multicast and anycast.
- IP addresses are divided into two parts: network prefix and host identifier. The network prefix identifies the network or subnet that the device belongs to, and the host identifier identifies the specific device within that network or subnet.
- The network prefix and host identifier are separated by a slash (/) followed by a number that indicates the length of the network prefix in bits. For example, 192.168.1.100/24 means that the network prefix is 192.168.1 and the host identifier is 100, and the network prefix is 24 bits long.
- IP addresses can be classified into different classes based on the length and range of the network prefix. For example, class A addresses have a

network prefix of 8 bits and a host identifier of 24 bits, and can support up to 126 networks and 16 million hosts per network. Class B addresses have a network prefix of 16 bits and a host identifier of 16 bits, and can support up to 16,384 networks and 65,536 hosts per network. Class C addresses have a network prefix of 24 bits and a host identifier of 8 bits, and can support up to 2 million networks and 254 hosts per network.

- IP addresses can also be categorized into different types based on their function and scope. For example, unicast addresses are used to identify a single device on the network, and are the most common type of IP addresses. Broadcast addresses are used to send data to all devices on the same network or subnet, and are usually the last address in the network or subnet range. Multicast addresses are used to send data to a group of devices that share a common interest, and are usually in the range of 224.0.0.0 to 239.255.255.255. Anycast addresses are used to send data to the nearest device that provides a certain service, and are usually assigned to multiple devices that have the same address.
- IP packets consist of two parts: header and payload. The header contains information such as the source and destination addresses, the protocol version, the length, the time to live, and the checksum. The payload contains the actual data that is being transmitted, such as an HTTP request or a DNS query.
- IP packets can be fragmented if they are larger than the maximum transmission unit (MTU) of the link layer protocol. Fragmentation is the process of breaking a large packet into smaller packets that can fit the MTU. Each fragment has its own header, which includes a fragment offset and a flag to indicate whether it is the last fragment or not. The fragments are reassembled at the destination device based on the fragment offset and the identification field in the header.
- IP routing is the process of finding the best path for a packet to reach its destination. IP routing is done by routers, which are devices that connect different networks and forward packets based on their destination addresses. Routers use routing tables, which are databases that store information about the networks and subnets that they are connected to, and the next hop router for each destination. Routers can learn routing information from static configuration, dynamic protocols, or both.
- IP supports several protocols that work at the transport layer (layer 4) of the OSI model, such as TCP, UDP, and ICMP. TCP (Transmission Control Protocol) is a reliable, connection-oriented, and stream-based protocol that provides error detection, flow control, and congestion control. UDP (User Datagram Protocol) is

CIDR

- CIDR stands for **Classless Inter-Domain Routing**, a method for allocating IP addresses and for IP routing.
- CIDR was introduced in 1993 to replace the previous classful network

addressing architecture on the Internet , which was based on fixed classes of IP addresses (A, B, and C).

- CIDR allows blocks of IP addresses to be grouped into single routing table entries, which reduces the size and complexity of routing tables on routers across the Internet .
- CIDR is based on a bitwise, prefix-based representation of IP addresses and their routing properties. A CIDR IP address looks like a normal IP address followed by a slash and a number, called the **network prefix**. For example, `192.168.1.0/24` is a CIDR IP address.
- The network prefix indicates how many bits of the IP address are used to identify the network, and how many bits are used to identify the host within the network. The network prefix can range from 0 to 32 for IPv4 addresses, and from 0 to 128 for IPv6 addresses.
- The network prefix can be used to calculate the **network address**, the **broadcast address**, and the **host range** of a CIDR block. The network address is the first IP address in the block, obtained by setting all the host bits to zero. The broadcast address is the last IP address in the block, obtained by setting all the host bits to one. The host range is the set of IP addresses between the network address and the broadcast address, excluding the network address and the broadcast address themselves.
- For example, the CIDR block `192.168.1.0/24` has a network address of `192.168.1.0`, a broadcast address of `192.168.1.255`, and a host range of `192.168.1.1` to `192.168.1.254`. It can accommodate up to 254 hosts in the network.
- CIDR allows more efficient and flexible use of IP addresses, as it can create networks of different sizes based on the network prefix. For example, a `/24` network can be divided into four `/26` networks, each with 62 hosts, or two `/25` networks, each with 126 hosts. Conversely, a `/24` network can be combined with another `/24` network to form a `/23` network, with 510 hosts.
- CIDR also enables **supernetting**, which is the process of aggregating multiple contiguous networks into a larger network. For example, four `/24` networks can be aggregated into a `/22` network, with 1022 hosts. Supernetting reduces the number of routing entries and improves routing efficiency.
- CIDR is an essential technique for managing the limited IP address space, especially for IPv4 addresses, which have been exhausted since 2011. CIDR also facilitates the transition to IPv6 addresses, which have a much larger address space and use a similar notation.

ARP

- ARP stands for Address Resolution Protocol. It is a network protocol that maps an IP address to a MAC address of a device on the same network.
- ARP operates at the data link layer (layer 2) of the OSI model. It uses

broadcast messages to request and reply the MAC address of a device with a given IP address.

- ARP is used when a device wants to communicate with another device on the same network, but does not know its MAC address. For example, when a computer wants to send a packet to a printer on the same LAN, it needs to know the printer's MAC address to encapsulate the packet in an Ethernet frame.
- The basic steps of ARP are:
 1. The sender broadcasts an ARP request message to all devices on the network, asking "Who has this IP address? Please send me your MAC address."
 2. The device with the matching IP address replies with an ARP reply message, containing its MAC address.
 3. The sender receives the ARP reply and updates its ARP cache, a table that stores the IP-MAC mappings of known devices.
 4. The sender can now send the packet to the destination using the MAC address from the ARP cache.
- An example of an ARP request and reply message is shown below:

ARP request:

Hardware type (16b)	Protocol type (16b)	Hardware size (8b)	Protocol size (8b)
Ethernet (0x0001)	IPv4 (0x0800)	6	4
Sender IP (32b)	Target MAC (48b)	Target IP (32b)	
192.168.1.10	00:00:00:00:00:00	192.168.1.20	

ARP reply:

Hardware type (16b)	Protocol type (16b)	Hardware size (8b)	Protocol size (8b)
Ethernet (0x0001)	IPv4 (0x0800)	6	4
Sender IP (32b)	Target MAC (48b)	Target IP (32b)	
192.168.1.20	aa:bb:cc:dd:ee:ff	192.168.1.10	

- Some advantages of ARP are:
 - It is simple and efficient, as it only requires two messages to resolve an IP address to a MAC address.

- It is dynamic, as it can update the ARP cache with the latest information of the devices on the network.
- It is flexible, as it can support different types of hardware and protocol addresses.
- Some disadvantages of ARP are:
 - It is vulnerable to spoofing attacks, where a malicious device can send fake ARP messages to trick other devices into believing that it has a certain IP or MAC address.
 - It can cause network congestion, as it uses broadcast messages that are sent to all devices on the network, regardless of their relevance.
 - It can waste resources, as it requires each device to maintain an ARP cache that may contain outdated or unnecessary entries.
- Some applications of ARP are:
 - It is essential for the communication between devices on the same network, such as LANs or VLANs.
 - It is used by other network protocols that rely on MAC addresses, such as RARP

RARP

- RARP stands for **Reverse Address Resolution Protocol** .
- It is a protocol based on computer networking that is used by a client computer to request its **IP address** from a gateway server's **ARP table** or cache .
- It works on the **Network Access Layer** (the lowest layer) of the TCP/IP protocol stack .
- It is useful for devices that do not have a permanent IP address, such as diskless workstations or printers .
- It operates by sending a **broadcast packet** containing the device's **MAC address** to the network, and waiting for a **RARP server** to reply with the corresponding IP address .
- A RARP server is a specialized device that has a table or database of MAC addresses and IP addresses, and can respond to RARP requests .
- The format of a RARP packet is similar to an ARP packet, except that the **operation code** is set to 3 for request and 4 for reply.
- A RARP packet has the following fields:

Field	Size (bytes)	Description
Hardware type	2	Specifies the type of network hardware, such as Ethernet
Protocol type	2	Specifies the type of network protocol, such as IP

Field	Size (bytes)	Description
Hardware address length	1	Specifies the length of the hardware address, such as 6 for MAC address
Protocol address length	1	Specifies the length of the protocol address, such as 4 for IP address
Operation code	2	Specifies the type of RARP message, such as 3 for request and 4 for reply
Sender hardware address	Variable	Specifies the MAC address of the sender
Sender protocol address	Variable	Specifies the IP address of the sender (0.0.0.0 for request)
Target hardware address	Variable	Specifies the MAC address of the target (same as sender for request)
Target protocol address	Variable	Specifies the IP address of the target (0.0.0.0 for request, assigned IP address for reply)

- An example of a RARP request and reply is shown below:

RARP request:

```
| Hardware type | Protocol type | Hardware address length | Protocol address length | Operat
| --- | --- | --- | --- | --- | --- | --- | --- | --- |
| 1 (Ethernet) | 0x0800 (IP) | 6 | 4 | 3 (request) | 00:0C:29:0A:0B:0C | 0.0.0.0 | 00:0C:29:0A:0B:0C
```

RARP reply:

```
| Hardware type | Protocol type | Hardware address length | Protocol address length | Operat
| --- | --- | --- | --- | --- | --- | --- | --- | --- |
| 1 (Ethernet) | 0x0800 (IP) | 6 | 4 | 4 (reply) | 00:0C:29:0A:0B:0D | 192.168.1.1 | 00:0C:29:0A:0B:0D
```

- RARP has some limitations and disadvantages, such as :
 - It requires a RARP server on the same LAN as the client, which may not be available or reliable.
 - It only works with IP addresses, and cannot support other network protocols.

- It only provides the IP address, and not other configuration information, such as subnet mask, default gateway, or DNS server.
- It uses broadcast packets, which may cause network congestion or security issues.
- RARP has been superseded by other protocols, such as ****BOOTP**

DHCP

- DHCP stands for Dynamic Host Configuration Protocol.
- It is a network protocol that assigns IP addresses and other network parameters to devices dynamically.
- It operates on the application layer of the TCP/IP model and uses UDP as the transport protocol.
- It works in four steps: Discover, Offer, Request, and Acknowledge (DORA).
 - Discover: The client broadcasts a DHCPDISCOVER message to find a DHCP server on the network.
 - Offer: The DHCP server responds with a DHCPOFFER message, offering an IP address and other network parameters to the client.
 - Request: The client accepts the offer and broadcasts a DHCPREQUEST message, requesting the IP address and other network parameters from the server.
 - Acknowledge: The DHCP server confirms the allocation and sends a DHCPACK message to the client, acknowledging the request.
- DHCP has several advantages, such as:
 - It simplifies network administration by automating IP address assignment and configuration.
 - It reduces network conflicts and errors by avoiding duplicate IP addresses and ensuring consistency of network parameters.
 - It supports network scalability and mobility by allowing devices to join and leave the network easily and dynamically.
- DHCP has some disadvantages, such as:
 - It introduces security risks by exposing network information and allowing unauthorized devices to obtain IP addresses and access the network.
 - It depends on the availability and reliability of the DHCP server, which may fail or be overloaded by requests.
 - It consumes network bandwidth and resources by generating broadcast messages and requiring periodic renewals of IP addresses.
- A mnemonic to remember the four steps of DHCP is: **Do Or Regret Action (DORA)**.
- An example of a DHCPDISCOVER message is:

```

+-----+
| OP: 1 (request) |
+-----+

```

```

| HTYPE: 1 (Ethernet) |
+-----+
| HLEN: 6              |
+-----+
| HOPS: 0              |
+-----+
| XID: 0x12345678      |
+-----+
| SECS: 0              |
+-----+
| FLAGS: 0x0000        |
+-----+
| CIADDR: 0.0.0.0      |
+-----+
| YIADDR: 0.0.0.0      |
+-----+
| SIADDR: 0.0.0.0      |
+-----+
| GIADDR: 0.0.0.0      |
+-----+
| CHADDR: 00:11:22:33:44:55 |
+-----+
| SNAME: (empty)       |
+-----+
| FILE: (empty)        |
+-----+
| OPTIONS:              |
| - 53: DHCP Message Type (1: Discover) |
| - 55: Parameter Request List (1: Subnet Mask, 3: Router, 6: DNS Server, 15: Domain Name)
| - 255: End            |
+-----+

```

- An example of a DHCPOFFER message is:

```

+-----+
| OP: 2 (reply)        |
+-----+
| HTYPE: 1 (Ethernet) |
+-----+
| HLEN: 6              |
+-----+
| HOPS: 0              |
+-----+
| XID: 0x12345678      |
+-----+
| SECS: 0              |
+-----+

```

```

| FLAGS: 0x0000      |
+-----+
| CIADDR: 0.0.0.0    |
+-----+
| YIADDR: 192.168.1.100 |
+-----+
| SIADDR: 192.168.1.1 |
+-----+
| GIADDR: 0.0.0.0    |
+-----+
| CHADDR: 00:11:22:33:44:55 |
+-----+
| SNAME: (empty)     |
+-----+
| FILE: (empty)      |
+-----+
| OPTIONS:           |
| - 53: DHCP Message Type (2: Offer) |
| - 1: Subnet Mask (255.255.255.0) |
| - 3: Router (192.168.1.1) |
| - 6: DNS Server (8.8.8.8) |
| - 15: Domain Name (example.com) |
| - 51: IP Address Lease Time (86400)

```

ICMP

- ICMP stands for Internet Control Message Protocol. It is a network layer protocol that is used by network devices to communicate problems with data transmission .
- ICMP messages are typically used for diagnostic or control purposes or generated in response to errors in IP operations (as specified in RFC 1122). For example, ICMP can be used to ping a host to check its availability, or to trace the route of a packet to its destination.
- ICMP messages have a common header format, followed by a variable-length data field. The header consists of four fields: type, code, checksum, and identifier. The type field specifies the kind of ICMP message, such as echo request, echo reply, destination unreachable, time exceeded, etc. The code field provides more details about the type of message. The checksum field is used to verify the integrity of the message. The identifier field is used to match the request and reply messages.
- ICMP messages are encapsulated within IP datagrams, and are therefore subject to the same routing and fragmentation rules as any other IP packet. However, ICMP messages are not reliable, meaning that they may be lost, delayed, or reordered during transmission. Therefore, ICMP messages should not be used to convey critical information or to implement reliable protocols.

- ICMP messages can be useful for troubleshooting network problems, such as connectivity issues, routing loops, packet loss, congestion, etc. However, ICMP messages can also be abused by attackers to launch denial-of-service (DoS) attacks, reconnaissance attacks, or to bypass firewalls. Therefore, network administrators should monitor and filter ICMP traffic carefully to prevent malicious use.

Some mnemonics and learning tricks for ICMP are:

- ICMP can be remembered as “I Can’t Manage Packets”, which reflects its role in reporting errors and problems with IP packets.
- The four fields in the ICMP header can be remembered as “Type, Code, Check, ID”, which can be pronounced as “Tick, Code, Check, ID”.
- The most common ICMP types can be remembered as “0, 3, 8, 11”, which correspond to echo reply, destination unreachable, echo request, and time exceeded, respectively. These can be pronounced as “Zero, Three, Eight, Eleven”.

Routing in network layer

- Routing is the process of finding a path from a source to a destination in a network of nodes, such as routers, switches, or hosts.
- Routing is performed by the network layer, which is responsible for delivering packets across different networks.
- Routing can be classified into two types: static routing and dynamic routing.
- Static routing is when the routes are manually configured by the network administrator and do not change unless updated manually.
- Dynamic routing is when the routes are automatically learned and updated by the routers using routing protocols, such as RIP, OSPF, EIGRP, BGP, etc.
- Routing protocols can be further classified into two types: distance vector and link state.
- Distance vector protocols are based on the principle of exchanging distance information (such as hop count or cost) with neighboring routers. Each router maintains a routing table that contains the best known distance to each destination and the next hop to reach it. Examples of distance vector protocols are RIP and EIGRP.
- Link state protocols are based on the principle of exchanging topology information (such as link status, bandwidth, delay, etc.) with all routers in the network. Each router maintains a link state database that contains the complete map of the network and uses a shortest path algorithm (such as Dijkstra’s or Bellman-Ford) to compute the best path to each destination. Examples of link state protocols are OSPF and IS-IS.
- Routing can also be classified into two types: unicast and multicast.
- Unicast routing is when a packet is sent from a single source to a single destination. Unicast routing can use any of the routing protocols mentioned

above.

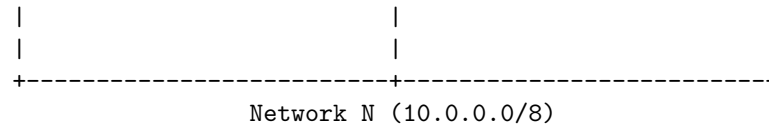
- Multicast routing is when a packet is sent from a single source to a group of destinations that share a common interest in the packet. Multicast routing requires special protocols that can build and maintain multicast trees, such as DVMRP, PIM, MOSPF, etc.

Forwarding and delivery in network layer

- Forwarding is the process of moving packets from an input interface to an output interface on a router based on the destination address and routing table.
- Delivery is the process of delivering packets to the final destination host or network.
- Forwarding and delivery are two main functions of the network layer in the TCP/IP model or the internet layer in the OSI model.
- Forwarding can be classified into two types: static and dynamic.
 - Static forwarding is when the routing table is manually configured by the network administrator and does not change unless updated manually.
 - Dynamic forwarding is when the routing table is automatically updated by routing protocols that exchange information with other routers and adapt to network changes.
- Delivery can be classified into two types: direct and indirect.
 - Direct delivery is when the source and destination hosts are on the same network and the packet is delivered directly to the destination host without passing through any routers.
 - Indirect delivery is when the source and destination hosts are on different networks and the packet is delivered through one or more routers until it reaches the destination network and then the destination host.
- A mnemonic to remember the difference between forwarding and delivery is: **F**orwarding is **F**rom input to output, **D**elivery is **D**estination host or network.
- A mnemonic to remember the difference between static and dynamic forwarding is: **S**tatic is **S**et by hand, **D**ynamic is **D**one by protocol.
- A mnemonic to remember the difference between direct and indirect delivery is: **D**irect is **D**one without router, **I**ndirect is **I**nvolving router.
- An example of forwarding and delivery in network layer is shown below:

Source host A (192.168.1.10/24) wants to send a packet to destination host B (10.0.0.20/8)

+-----+	+-----+	+-----+
Host A	Router R1	Host B
IP: 192.168.1.10/24	IP: 192.168.1.1/24	IP: 10.0.0.20/8
	10.0.0.1/8	
+-----+	+-----+	+-----+



- Host A performs direct delivery to router R1, since they are on the same network (192.168.1.0/24).
- Host A sets the destination IP address as 10.0.0.20 and the destination MAC address as the MAC address of router R1.
- Host A sends the packet to router R1 using the data link layer protocol (e.g. Ethernet).
- Router R1 performs forwarding to host B, since it has an interface on the destination network (10.0.0.0/8).
- Router R1 looks up the destination IP address in its routing table and finds the output interface as the interface connected to the destination network.
- Router R1 sets the source IP address as 10.0.0.1 and the source MAC address as its own MAC address.
- Router R1 sets the destination IP address as 10.0.0.20 and the destination MAC address as the MAC address of host B.
- Router R1 sends the packet to host B using the data link layer protocol (e.g. Ethernet).
- Host B performs direct delivery to itself, since it is the final destination host.
- Host B receives the packet and checks the destination IP address and MAC address.
- Host B processes the packet at the network layer and above.

Static and dynamic routing in computer networks

- Static routing and dynamic routing are two methods of configuring routing tables in a network.
- Routing tables are data structures that store information about the best paths to reach different destinations in a network.
- Static routing means that the routing tables are manually configured by the network administrator and do not change unless the administrator updates them.
- Dynamic routing means that the routing tables are automatically updated by routing protocols that exchange information with other routers in the network.
- Some advantages of static routing are:
 - It is simple and easy to implement in small networks.
 - It is more secure and predictable than dynamic routing, as the administrator has full control over the routes.
 - It consumes less bandwidth and CPU resources than dynamic routing, as there is no routing protocol overhead.
- Some disadvantages of static routing are:
 - It is difficult and time-consuming to implement and maintain in large and complex networks.
 - It is not scalable or adaptable to network changes, such as link failures, congestion, or topology changes.
 - It can lead to suboptimal routing and performance issues, as the administrator may not have complete or accurate knowledge of the network topology and conditions.
- Some advantages of dynamic routing are:

- It is easier and faster to implement and maintain in large and dynamic networks, as the routing tables are automatically updated by the routing protocols.
- It is more scalable and adaptable to network changes, as the routing protocols can discover new routes, avoid failed links, and balance traffic load.
- It can lead to optimal routing and performance, as the routing protocols can select the best paths based on various metrics, such as hop count, bandwidth, delay, or cost.
- Some disadvantages of dynamic routing are:
 - It is more complex and difficult to implement in small and simple networks, as the routing protocols require configuration and tuning of various parameters and options.
 - It is less secure and predictable than static routing, as the routing tables can be manipulated or compromised by malicious or faulty routers.
 - It consumes more bandwidth and CPU resources than static routing, as the routing protocols generate and exchange routing information periodically or on demand.
- Some examples of static routing are:
 - Default routes, which are used to direct packets to a default gateway when no specific route is available for the destination.
 - Host routes, which are used to direct packets to a specific host or device in the network.
 - Network routes, which are used to direct packets to a specific network or subnet in the network.
- Some examples of dynamic routing are:
 - Distance vector routing protocols, such as RIP and EIGRP, which use the hop count as the metric and exchange routing information with their immediate neighbors.
 - Link state routing protocols, such as OSPF and IS-IS, which use the shortest path algorithm and exchange routing information with all routers in the same area or domain.
 - Path vector routing protocols, such as BGP, which use the path attributes and exchange routing information with selected routers in different autonomous systems.

Routing algorithms and protocols in cn

- Routing is the process of finding a path from a source to a destination in a network of nodes.
- Routing algorithms are the methods or rules that determine how to select the best path for a packet to travel from the source to the destination.
- Routing protocols are the set of rules or standards that define how routers communicate with each other to exchange routing information and maintain routing tables.

- There are two main types of routing algorithms: static and dynamic.
 - Static routing algorithms use fixed or predefined routes that do not change unless manually configured by the network administrator. Static routing is simple, fast, and secure, but it is not adaptive to network changes or failures, and it requires manual intervention to update the routes.
 - Dynamic routing algorithms use current network conditions, such as traffic load, link status, or topology changes, to update the routes automatically. Dynamic routing is adaptive, flexible, and scalable, but it is more complex, slower, and less secure than static routing, and it requires more network resources and overhead to exchange routing information.
- There are two main types of routing protocols: distance vector and link state.
 - Distance vector routing protocols use the distance or hop count as the metric to select the best path. Each router maintains a routing table that contains the distance and the next hop to reach each destination. The routers exchange their routing tables periodically with their neighbors, and update their own tables based on the information received. Distance vector routing protocols are simple, easy to implement, and suitable for small networks, but they are slow to converge, prone to routing loops, and do not consider other factors such as bandwidth or delay.
 - Link state routing protocols use the link cost as the metric to select the best path. Each router maintains a link state database that contains the status and the cost of each link in the network. The routers exchange link state packets with all other routers in the network, and use a shortest path algorithm, such as Dijkstra's algorithm, to compute the best path to each destination. Link state routing protocols are fast to converge, loop-free, and consider multiple factors, but they are complex, difficult to implement, and require more network resources and overhead to exchange link state packets.
- Some examples of routing algorithms and protocols are:
 - RIP (Routing Information Protocol): a distance vector routing protocol that uses hop count as the metric and has a maximum hop count of 15. RIP is simple and widely used, but it has slow convergence, routing loops, and poor scalability.
 - OSPF (Open Shortest Path First): a link state routing protocol that uses link cost as the metric and supports multiple areas and hierarchical routing. OSPF is fast, loop-free, and scalable, but it is complex and requires more network resources.
 - EIGRP (Enhanced Interior Gateway Routing Protocol): a hybrid routing protocol that combines the features of distance vector and link state routing protocols. EIGRP uses a composite metric that includes bandwidth, delay, reliability, and load, and supports unequal cost load balancing. EIGRP is fast, loop-free, and scalable, but it is

- BGP (Border Gateway Protocol): a path vector routing protocol that is used to exchange routing information between autonomous systems (AS) on the Internet. BGP uses attributes, such as AS path, next hop, local preference, and origin, to select the best path. BGP is flexible, robust, and scalable, but it is complex and requires manual configuration.

Congestion control algorithms in cn

Congestion control algorithms are techniques to prevent or reduce network congestion by regulating the amount of data that can be sent by the source nodes. Congestion occurs when the demand for network resources exceeds the available capacity, resulting in packet loss, delay, and reduced throughput.

Some of the common congestion control algorithms are:

- **Additive Increase Multiplicative Decrease (AIMD):** This algorithm increases the window size by one segment for every successful transmission (additive increase) and halves the window size for every packet loss (multiplicative decrease). This algorithm is fair and stable, but it is slow to converge to the optimal window size and it may cause oscillations around the equilibrium point.
- **Multiplicative Increase Multiplicative Decrease (MIMD):** This algorithm multiplies the window size by a constant factor for every successful transmission (multiplicative increase) and multiplies the window size by another constant factor for every packet loss (multiplicative decrease). This algorithm is faster to converge to the optimal window size, but it is unfair and unstable, as it may cause some flows to dominate the network and others to starve.
- **Additive Increase Additive Decrease (AIAD):** This algorithm increases the window size by a constant amount for every successful transmission (additive increase) and decreases the window size by a constant amount for every packet loss (additive decrease). This algorithm is fair and stable, but it is slow to converge to the optimal window size and it may cause underutilization of the network.
- **Multiplicative Increase Additive Decrease (MIAD):** This algorithm multiplies the window size by a constant factor for every successful transmission (multiplicative increase) and decreases the window size by a constant amount for every packet loss (additive decrease). This algorithm is faster to converge to the optimal window size, but it is unfair and unstable, as it may cause some flows to dominate the network and others to starve.
- **Binary Exponential Backoff (BEB):** This algorithm is used in Ethernet networks to resolve collisions. It works by doubling the random

waiting time after each collision, until a maximum value is reached or the transmission is successful. This algorithm reduces the probability of collisions, but it may cause long delays and waste bandwidth.

- **Slow Start:** This algorithm is used in TCP to avoid sending too many packets at the beginning of a connection. It works by starting with a small window size and doubling it for every successful transmission, until a threshold value is reached or a packet loss occurs. This algorithm allows the sender to probe the network capacity and avoid congestion, but it may cause slow transmission rates for short-lived connections.
- **Congestion Avoidance:** This algorithm is used in TCP to avoid congestion after the slow start phase. It works by increasing the window size by one segment for every round trip time (RTT), until a packet loss occurs. Then, it reduces the window size by half and enters the slow start phase again. This algorithm maintains a balance between efficiency and fairness, but it may cause oscillations around the optimal window size.
- **Fast Retransmit:** This algorithm is used in TCP to detect and recover from packet losses. It works by using duplicate acknowledgments (ACKs) to infer packet losses. If the sender receives three duplicate ACKs for the same segment, it assumes that the segment was lost and retransmits it without waiting for a timeout. This algorithm reduces the delay and improves the throughput, but it may not work well for multiple packet losses or out-of-order segments.
- **Fast Recovery:** This algorithm is used in TCP to avoid reducing the window size too much after a packet loss. It works by using the duplicate ACKs to estimate the new window size, instead of halving it. The sender sets the window size to the number of outstanding segments plus the number of duplicate ACKs, and continues the congestion avoidance phase. This algorithm avoids the slow start phase and maintains a high transmission rate, but it may cause unfairness and instability in some scenarios.

IPv6 in cn

- IPv6 stands for Internet Protocol version 6, which is the latest version of the Internet Protocol (IP) that is used to identify and communicate with devices on the Internet.
- IPv6 was developed to overcome the limitations of IPv4, such as the exhaustion of address space, the lack of security features, the complexity of configuration and management, and the poor support for mobility and quality of service (QoS).
- IPv6 has the following features and advantages over IPv4:
 - Larger address space: IPv6 uses 128-bit addresses, which can support 2^{128} (approximately 3.4×10^{38}) possible addresses, compared to 32-bit addresses in IPv4, which can support 2^{32} (approximately

4.3×10^9) possible addresses. This allows for more devices to be connected to the Internet, and for more efficient and hierarchical address allocation.

- Stateless address autoconfiguration (SLAAC): IPv6 enables devices to automatically configure their own addresses without the need for a server, such as a Dynamic Host Configuration Protocol (DHCP) server. This simplifies the network configuration and management, and reduces the overhead and latency of address assignment.
 - Enhanced security: IPv6 supports the mandatory use of the IP Security (IPsec) protocol, which provides encryption and authentication for the data packets. This improves the confidentiality, integrity, and availability of the communication, and prevents attacks such as spoofing, replay, and denial of service (DoS).
 - Improved performance: IPv6 has a simplified header format, which reduces the processing and routing overhead of the packets. IPv6 also supports extension headers, which allow for the inclusion of optional information, such as routing, fragmentation, mobility, and QoS. IPv6 also eliminates the need for network address translation (NAT), which can cause problems for some applications and protocols, such as peer-to-peer (P2P) and Voice over IP (VoIP).
 - Support for mobility: IPv6 supports the Mobile IPv6 (MIPv6) protocol, which enables devices to move from one network to another without changing their addresses or losing their connections. This enhances the user experience and the efficiency of the network resources.
 - Support for QoS: IPv6 supports the Differentiated Services (DiffServ) and Integrated Services (IntServ) models, which enable the prioritization and reservation of network resources for different types of traffic, such as real-time, interactive, and bulk data. This improves the quality and reliability of the communication, and reduces the delay and jitter of the packets.
- A mnemonic to remember the IPv6 address format is: **8 groups of 4 hex digits separated by colons**. For example, 2001:0db8:85a3:0000:0000:8a2e:0370:7334 is a valid IPv6 address.
 - A learning trick to convert between binary and hexadecimal is to **group the binary digits into four and map them to the corresponding hexadecimal digit**. For example, 1010 0001 1100 1111 in binary is equivalent to A1CF in hexadecimal.

Unit 4 - Transport Layer in Computer Networks

The transport layer is a conceptual division of methods in the layered architecture of protocols in the network stack in the Internet protocol suite and the OSI model. The protocols of this layer provide end-to-end communication services for applications.

Some of the main functions and features of the transport layer are:

- **Segmentation and reassembly:** The transport layer takes data from the upper layer (i.e. application layer) and then breaks it into smaller size segments, numbers each byte, and hands over to the lower layer (network layer) for delivery. At the receiver side, the transport layer reassembles the segments into the original data.
- **Addressing:** The transport layer provides the user address which is specified as a station or port. The port variable represents a specific process or service running on a host. The transport layer protocols need to know the port number of the destination process to deliver the data correctly.
- **Multiplexing and demultiplexing:** The transport layer can multiplex multiple applications or processes on the same host to use the same network layer connection by assigning different port numbers to each process. Similarly, the transport layer can demultiplex the incoming data from the network layer to the appropriate application or process by using the port numbers.
- **Reliable and unreliable delivery:** The transport layer can provide different types of delivery services depending on the requirements of the applications or processes. Some transport layer protocols, such as TCP, provide reliable delivery by using error detection, acknowledgment, retransmission, and flow control mechanisms. Other transport layer protocols, such as UDP, provide unreliable delivery by sending the data without any guarantee of delivery or error recovery.
- **Connection-oriented and connectionless communication:** The transport layer can support both connection-oriented and connectionless communication modes. In connection-oriented communication, the transport layer establishes a logical connection between the source and destination hosts before sending any data, and terminates the connection after the data transfer is complete. In connectionless communication, the transport layer sends the data without any prior connection establishment or termination.
- **Quality of service:** The transport layer can provide different levels of quality of service (QoS) to the applications or processes by using parameters such as bandwidth, delay, jitter, and loss. The transport layer can negotiate the QoS requirements with the network layer or the application layer, and adapt the delivery service accordingly.

Some of the common transport layer protocols are:

- **Transmission Control Protocol (TCP):** TCP is a connection-oriented, reliable, byte-stream protocol that provides error detection, acknowledgment, retransmission, and flow control mechanisms. TCP is widely used for applications that require reliable and ordered delivery of data, such as web browsing, email, file transfer, etc.
- **User Datagram Protocol (UDP):** UDP is a connectionless, unreliable, datagram protocol that provides no error detection, acknowledgment, re-

transmission, or flow control mechanisms. UDP is widely used for applications that require fast and efficient delivery of data, such as video streaming, voice over IP, online gaming, etc.

- **Stream Control Transmission Protocol (SCTP)**: SCTP is a connection-oriented, reliable, message-oriented protocol that provides error detection, acknowledgment, retransmission, and flow control mechanisms. SCTP also supports multiple streams of data within a single connection, and can handle network failures by switching to alternate paths. SCTP is widely used for applications that require high availability and reliability, such as telecommunication signaling, web conferencing, etc.

Some of the mnemonics and learning tricks for the transport layer are:

- To remember the main functions of the transport layer, use the acronym **SARMRQ** (Segmentation, Addressing, Multiplexing, Reliable/Unreliable delivery, QoS).
- To remember the difference between TCP and UDP, use the analogy of **phone call vs. postcard**. A phone call is like TCP, which requires a connection, guarantees delivery, and maintains order. A postcard is like UDP, which does not require a connection, does not guarantee delivery, and does not maintain order.
- To remember the difference between connection-oriented and connectionless communication, use the analogy of **train vs. bus**. A train is like connection-oriented communication, which requires a reservation, follows a fixed route, and arrives at a specific destination. A bus is like connectionless communication, which does not require a reservation, can take different routes, and can stop at any station.

Process-to-process delivery in transport layer

- The transport layer is responsible for delivering data from one process to another process on different or same hosts.
- A process is an instance of a running program that can send or receive data.
- A process is identified by a port number, which is a 16-bit integer that ranges from 0 to 65535.
- The port number is appended to the IP address of the host to form a socket address, which uniquely identifies a process on a network.
- The transport layer uses two protocols to provide process-to-process delivery: TCP and UDP.
- TCP (Transmission Control Protocol) is a connection-oriented, reliable, and full-duplex protocol that provides flow control, congestion control, and error control.
- UDP (User Datagram Protocol) is a connectionless, unreliable, and simple protocol that does not provide any control mechanisms.
- TCP and UDP use header fields to carry information such as source port,

destination port, sequence number, acknowledgment number, checksum, etc.

- TCP and UDP also use multiplexing and demultiplexing techniques to allow multiple processes to share the same transport layer protocol.
- Multiplexing is the process of combining data from multiple processes into a single segment with a header that contains the port numbers of the source and destination processes.
- Demultiplexing is the process of extracting data from a single segment and delivering it to the appropriate process based on the port numbers in the header.
- A mnemonic to remember the difference between TCP and UDP is: TCP is Tight Control Protocol, UDP is Unreliable Datagram Protocol.

Transport layer protocols

- Transport layer protocols are responsible for providing end-to-end communication services for applications over a network.
- Transport layer protocols lie between the application layer and the network layer in the protocol stack.
- Transport layer protocols can be classified into two types: connection-oriented and connectionless.
- Connection-oriented protocols establish a logical connection between the sender and the receiver before exchanging data, and terminate the connection after the data transfer is complete. They also provide reliable data delivery, flow control, congestion control, and error recovery mechanisms. An example of a connection-oriented protocol is the Transmission Control Protocol (TCP).
- Connectionless protocols do not establish or terminate connections, and do not guarantee reliable data delivery, flow control, congestion control, or error recovery. They simply send datagrams (packets) from the sender to the receiver, without any acknowledgement or feedback. An example of a connectionless protocol is the User Datagram Protocol (UDP).
- Transport layer protocols identify applications communicating with each other by means of port numbers. Port numbers are 16-bit numbers that are assigned to specific applications or services by the Internet Assigned Numbers Authority (IANA). Port numbers 1 to 1023 are well known and reserved for standardized services, such as HTTP (80), FTP (21), and SSH (22). Port numbers 1024 to 49151 are registered and used by various applications, such as Skype (1024), BitTorrent (6881), and Minecraft (25565). Port numbers 49152 to 65535 are dynamic and used for temporary or ephemeral connections.
- Transport layer protocols also provide segmentation and reassembly of data. Segmentation is the process of dividing a large data stream into

smaller segments that can fit into the network layer packets. Reassembly is the process of reconstructing the original data stream from the received segments at the destination. Segmentation and reassembly help to reduce the network congestion and improve the efficiency of data transmission.

- Some of the transport layer protocols that are defined and implemented in the Internet protocol suite are:
 - TCP: Transmission Control Protocol is a connection-oriented, reliable, and full-duplex protocol that provides byte-stream service to the applications. TCP uses a three-way handshake to establish and terminate connections, and uses sequence numbers, acknowledgements, timers, and retransmission to ensure reliable data delivery. TCP also uses sliding window mechanism to provide flow control and congestion control. TCP is widely used for applications that require reliable and ordered data delivery, such as web browsing, email, file transfer, and remote login.
 - UDP: User Datagram Protocol is a connectionless, unreliable, and best-effort protocol that provides datagram service to the applications. UDP does not use any handshake, sequence numbers, acknowledgements, timers, or retransmission to establish or terminate connections, or to ensure reliable data delivery. UDP also does not provide any flow control or congestion control. UDP is widely used for applications that require fast and real-time data delivery, such as voice over IP, video streaming, online gaming, and DNS queries.
 - DCCP: Datagram Congestion Control Protocol is a connection-oriented, unreliable, and congestion-controlled protocol that provides datagram service to the applications. DCCP uses a four-way handshake to establish and terminate connections, and uses sequence numbers and acknowledgements to detect packet loss and provide feedback. DCCP also uses various congestion control mechanisms to adapt to the network conditions and avoid congestion collapse. DCCP is designed for applications that can tolerate some packet loss, but require congestion control, such as multimedia streaming, telephony, and online games.
 - SCTP: Stream Control Transmission Protocol is a connection-oriented, reliable, and message-oriented protocol that provides multiple streams of data to the applications. SCTP uses a four-way handshake to establish and terminate connections, and uses sequence numbers, acknowledgements, timers, and retransmission to ensure reliable data delivery. SCTP also uses sliding window mechanism to provide flow control and congestion control. SCTP supports multiple streams of data within a single connection, which allows for parallel and independent transmission of messages. SCTP also supports multihoming, which allows for multiple network paths between the endpoints. SCTP is designed for applications that require reliable and ordered delivery of messages, such as telecommunication

signaling, web services, and file transfer.

UDP Transport layer protocol

- UDP stands for User Datagram Protocol. It is one of the core protocols of the Internet protocol suite, along with TCP.
- UDP is a connectionless and unreliable protocol that provides datagram service to the application layer. It does not guarantee delivery, ordering, or error checking of the data packets.
- UDP is suitable for applications that require speed, efficiency, or real-time communication, such as streaming media, online gaming, voice over IP, or DNS queries.
- UDP has a simple header format that consists of four fields: source port, destination port, length, and checksum. The source and destination ports identify the endpoints of the communication, the length specifies the size of the UDP datagram in bytes, and the checksum is used for error detection.
- UDP does not provide any flow control, congestion control, or retransmission mechanisms. It relies on the application layer to handle these issues if needed.
- UDP is stateless, meaning that it does not keep track of the state of the communication or the status of the data packets. Each UDP datagram is treated independently by the protocol.
- UDP has a maximum payload size of 65,507 bytes, which is derived from the maximum IP datagram size of 65,535 bytes minus the 8-byte UDP header and the 20-byte IP header.
- UDP can be used to implement multicast and broadcast communication, which are not supported by TCP. Multicast allows sending data to multiple destinations simultaneously, while broadcast allows sending data to all hosts on a network segment.
- UDP is faster and more efficient than TCP, but less reliable and robust. UDP is preferred when the application can tolerate some data loss or delay, or when the application can handle the reliability issues itself.
- A mnemonic to remember the features of UDP is: **U**nreliable, **D**atagram, **P**ort.

Some possible additional sentences are:

- To use UDP, an application must create a socket and bind it to a port number. Then, it can send and receive UDP datagrams using the `sendto()` and `recvfrom()` functions.
- A UDP datagram can be encapsulated inside an IP datagram, which can be further encapsulated inside an Ethernet frame, for transmission over the network.
- A UDP datagram can be fragmented and reassembled by the IP layer if its size exceeds the maximum transmission unit (MTU) of the underlying network.

- A UDP datagram can be discarded by the network routers or the destination host if there is congestion, buffer overflow, or checksum error.

TCP Transport layer protocol

- TCP stands for Transmission Control Protocol. It is a transport layer protocol that facilitates the transmission of packets from source to destination.
- TCP is a connection-oriented protocol that means it establishes the connection prior to the communication that occurs between the computing devices in a network.
- TCP is a reliable protocol as it follows the flow and error control mechanism. It also supports the acknowledgment mechanism, which checks the state and sound arrival of the data.
- TCP provides reliable, ordered, and error-checked delivery of a stream of octets (bytes) between applications running on hosts communicating via an IP network.
- TCP uses a three-way handshake to establish a connection between two endpoints. The three steps are: SYN, SYN-ACK, and ACK.
- TCP uses a sliding window mechanism to control the flow of data and avoid congestion. The window size indicates how many bytes can be sent before waiting for an acknowledgment.
- TCP uses sequence numbers and acknowledgment numbers to keep track of the data sent and received. Each byte of data has a unique sequence number, and each acknowledgment contains the next expected sequence number.
- TCP uses checksums to detect errors in the data. The checksum is calculated based on the data and a pseudo-header that contains some information from the IP header. The checksum is verified at the receiver side.
- TCP can handle lost, duplicated, or out-of-order packets by using retransmission, selective acknowledgment, and sequence numbers.
- TCP can adjust the transmission rate based on the network conditions and the receiver's feedback. This is called congestion control. TCP uses various algorithms to implement congestion control, such as slow start, congestion avoidance, fast retransmit, and fast recovery.
- TCP can terminate a connection by using a four-way handshake. The four steps are: FIN, ACK, FIN, ACK.
- TCP can support multiple connections between the same pair of hosts by using port numbers. A port number is a 16-bit number that identifies a specific application or process on a host. A TCP connection is identified by a four-tuple: source IP address, source port number, destination IP address, and destination port number.
- TCP can support various options to enhance its functionality, such as maximum segment size, window scaling, timestamp, selective acknowledgment, and urgent pointer.

Some mnemonics and learning tricks for TCP are:

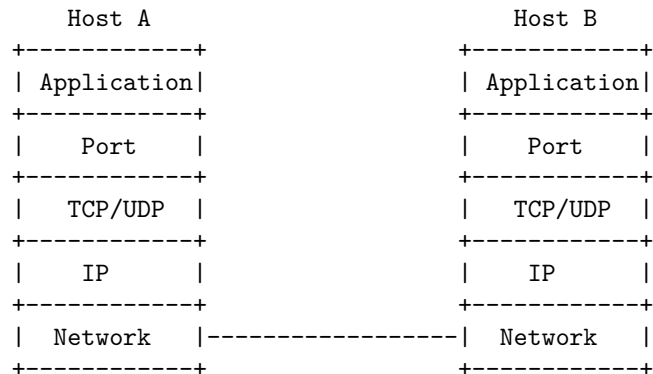
- To remember the three-way handshake, think of SYN as “synchronize”, SYN-ACK as “synchronize and acknowledge”, and ACK as “acknowledge”.
- To remember the four-way handshake, think of FIN as “finish”, and ACK as “acknowledge”.
- To remember the port numbers of some common applications, think of:
 - HTTP: 80 (eighty sounds like “hatey”, which is similar to “HTTP”)
 - HTTPS: 443 (four for secure, four for more, three for free)
 - FTP: 20 and 21 (file transfer protocol has two ports, one for data and one for control)
 - SSH: 22 (secure shell has two twos, which look like snakes)
 - Telnet: 23 (telnet has two threes, which look like phones)
 - SMTP: 25 (simple mail transfer protocol has two fives, which look like envelopes)
 - DNS: 53 (domain name system has five and three, which are the first and last letters of “DNS”)
 - DHCP: 67 and 68 (dynamic host configuration protocol has six and seven, which are the first and last letters of “DHCP”)
 - POP3: 110 (post office protocol version 3 has one and two zeros, which look like a mailbox)
 - IMAP: 143 (internet message access protocol has one and four threes, which look like letters)
 - SNMP: 161 (simple network management protocol has one and six ones, which look like sticks)

Multiplexing in transport layer

- Multiplexing is the process of combining multiple data streams from different sources into a single data stream for transmission over a shared medium.
- Multiplexing in transport layer is done by using port numbers to identify the source and destination applications of each data stream.
- Port numbers are 16-bit integers that range from 0 to 65535. They are divided into three categories: well-known ports (0-1023), registered ports (1024-49151), and dynamic or private ports (49152-65535).
- Well-known ports are assigned by the Internet Assigned Numbers Authority (IANA) to standard protocols and services, such as HTTP (80), FTP (21), and SSH (22).
- Registered ports are allocated by IANA to specific applications or organizations, such as Skype (49175), Minecraft (25565), and Apache Tomcat (8080).
- Dynamic or private ports are used by applications for temporary or ephemeral communication, such as web browsers and email clients. They are randomly chosen by the operating system from the available range.

- Multiplexing in transport layer allows multiple applications to share the same network interface and IP address, without interfering with each other's data.
- Multiplexing in transport layer also enables end-to-end communication between applications across different hosts, by using a combination of IP address and port number to uniquely identify each data stream.
- Multiplexing in transport layer is supported by both TCP and UDP protocols, but they use different methods to perform it.
- TCP multiplexing is based on the concept of a socket, which is a combination of an IP address, a port number, and a protocol. A socket uniquely identifies an endpoint of a TCP connection. TCP uses a four-tuple of (source IP, source port, destination IP, destination port) to multiplex and demultiplex data streams.
- UDP multiplexing is based on the concept of a datagram, which is a self-contained unit of data that contains an IP header, a UDP header, and a payload. A datagram does not establish a connection or maintain a state with the destination. UDP uses a two-tuple of (destination IP, destination port) to multiplex and demultiplex data streams.

A simple ASCII diagram to illustrate multiplexing in transport layer is:



Some mnemonics and learning tricks for multiplexing in transport layer are:

- Remember that multiplexing is like a **multi**-tasking **plex** movie theater, where different movies are shown on different screens to different audiences.
- Remember that port numbers are like **door** numbers of different rooms in a building, where different applications are running in different rooms.
- Remember that well-known ports are like **well-known** celebrities, who have fixed and famous names or numbers, such as Tom Cruise (80) or Brad Pitt (21).
- Remember that registered ports are like **registered** trademarks, who have specific and official names or numbers, such as Nike (49175) or Lego (25565).
- Remember that dynamic or private ports are like **dynamic** or **private**

individuals, who have random and temporary names or numbers, such as Alice (54321) or Bob (12345).

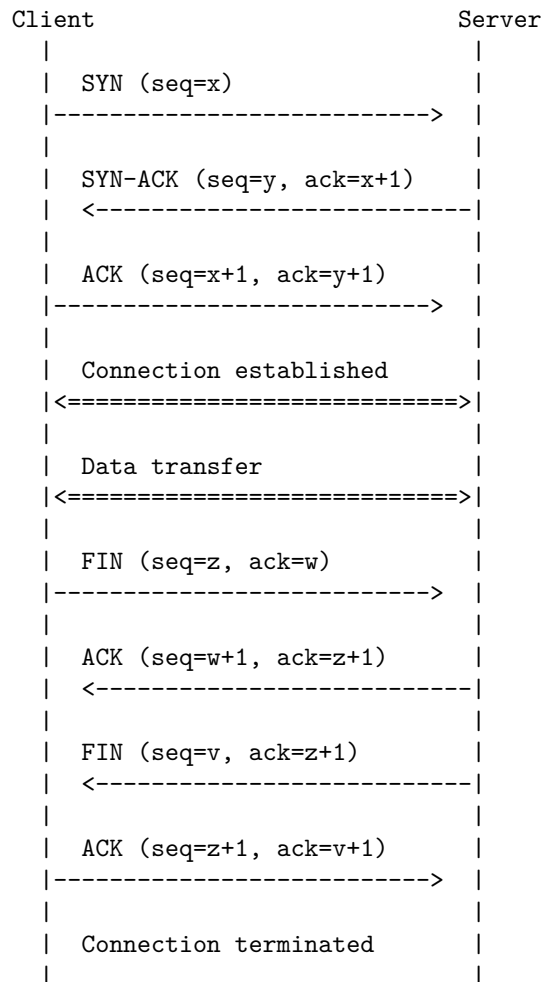
- Remember that TCP multiplexing is like a **TC**Phone call, where you need to dial a four-digit number (source IP, source port, destination IP, destination port) to connect to another person.
- Remember that UDP multiplexing is like a **UD**eliver**P**izza service, where you only need to give a two-digit number (destination IP, destination port) to send a pizza to another person.

Connection management in transport layer

- Connection management is the process of establishing, maintaining, and terminating a logical connection between two or more end points in a network.
- Connection management is one of the responsibilities of the transport layer, which provides end-to-end communication services for applications.
- Connection management can be classified into two types: connection-oriented and connectionless.
- Connection-oriented connection management requires the establishment of a connection before any data can be exchanged, and the termination of the connection after the data transfer is completed. Connection-oriented connection management ensures reliable, ordered, and error-free delivery of data, but it also introduces overhead and delay in the communication.
- Connectionless connection management does not require the establishment or termination of a connection, and data can be exchanged at any time without any prior agreement. Connectionless connection management is faster and more efficient, but it does not guarantee reliable, ordered, or error-free delivery of data.
- The most common protocols used for connection management in the transport layer are TCP (Transmission Control Protocol) and UDP (User Datagram Protocol).
- TCP is a connection-oriented protocol that uses a three-way handshake to establish a connection, and a four-way handshake to terminate a connection. TCP also uses sequence numbers, acknowledgments, timers, and retransmission mechanisms to ensure reliable data transfer. TCP is used for applications that require high reliability and accuracy, such as web browsing, email, file transfer, etc.
- UDP is a connectionless protocol that does not use any handshake or termination procedures, and does not provide any reliability or error control mechanisms. UDP simply sends datagrams to the destination without any guarantee of delivery, order, or integrity. UDP is used for applications

that require low latency and high efficiency, such as video streaming, voice over IP, online gaming, etc.

- A possible mnemonic to remember the difference between TCP and UDP is:
 - TCP: Trustworthy, Careful, and Polite
 - UDP: Unreliable, Daring, and Pragmatic
- A possible ascii diagram to illustrate the connection management in TCP is:



Flow control in transport layer

- Flow control is a mechanism that regulates the amount of data that can be sent by a sender to a receiver at a given time.

- Flow control is needed in transport layer because it prevents data loss due to congestion, buffer overflow, or mismatched speeds between the sender and the receiver.
- Flow control can be implemented by using feedback messages, such as acknowledgments, window sizes, or congestion signals, that inform the sender about the status of the receiver and the network.
- Flow control can be classified into two types: stop-and-wait and sliding window.
- Stop-and-wait flow control is a simple method that allows the sender to send one data unit at a time and wait for an acknowledgment from the receiver before sending the next one. This method ensures reliable delivery but has low efficiency and high latency.
- Sliding window flow control is a more advanced method that allows the sender to send multiple data units without waiting for acknowledgments, as long as the number of unacknowledged data units does not exceed a certain limit, called the window size. This method improves the efficiency and reduces the latency, but requires more complex logic and buffer management.
- An example of a transport layer protocol that uses stop-and-wait flow control is User Datagram Protocol (UDP), which is a connectionless and unreliable protocol that does not guarantee delivery, order, or error correction of data units.
- An example of a transport layer protocol that uses sliding window flow control is Transmission Control Protocol (TCP), which is a connection-oriented and reliable protocol that ensures delivery, order, and error correction of data units. TCP also adapts its window size dynamically based on the network conditions and the receiver's feedback, using algorithms such as slow start, congestion avoidance, and fast recovery.

etransmission in transport layer

- The transport layer is the fourth layer in the OSI model and the third layer in the TCP/IP model. It provides end-to-end communication services for applications, such as reliable data transfer, flow control, congestion control, error control, and multiplexing.
- etransmission is the process of sending and receiving data segments between the transport layer entities at the source and destination hosts. It involves the following steps:
 - The transport layer at the source host receives data from the application layer and divides it into segments. Each segment has a header that contains information such as source and destination port numbers, sequence numbers, acknowledgment numbers, checksum, and flags.
 - The transport layer at the source host sends the segments to the network layer, which encapsulates them into packets and forwards them to the destination host through the underlying network.

- The transport layer at the destination host receives the packets from the network layer and extracts the segments from them. It checks the checksum and the flags to verify the integrity and validity of the segments. It also checks the sequence numbers and acknowledgment numbers to detect any lost, duplicated, or out-of-order segments.
- The transport layer at the destination host sends acknowledgments (ACKs) to the source host for the segments that it has received correctly. It may also send negative acknowledgments (NACKs) or duplicate ACKs for the segments that it has not received or received incorrectly.
- The transport layer at the source host receives the ACKs, NACKs, or duplicate ACKs from the destination host and updates its window size, retransmission timer, and congestion window accordingly. It retransmits the segments that have not been acknowledged or have been negatively acknowledged. It also adjusts its sending rate to avoid or alleviate congestion in the network.
- The transport layer at the destination host reassembles the segments into the original data and delivers it to the application layer.
- The transport layer can use different protocols to perform transmission, such as TCP (Transmission Control Protocol) or UDP (User Datagram Protocol). TCP is a connection-oriented, reliable, and full-duplex protocol that provides error control, flow control, and congestion control. UDP is a connectionless, unreliable, and simplex protocol that does not provide any of these services. TCP is suitable for applications that require high reliability and low latency, such as web browsing, email, and file transfer. UDP is suitable for applications that can tolerate some loss and delay, such as video streaming, voice over IP, and online gaming.

Window management in transport layer

- Window management is a technique used by the transport layer to control the flow of data between two end systems.
- Window management ensures that the sender does not overwhelm the receiver or the network with too many packets at once, and that the receiver can acknowledge the received packets in a timely manner.
- Window management also helps to improve the efficiency and reliability of data transmission by detecting and recovering from packet losses, errors, or delays.
- Window management can be implemented using two main methods: **stop-and-wait** and **sliding window**.

Stop-and-wait

- In stop-and-wait, the sender sends one packet at a time and waits for an acknowledgment from the receiver before sending the next packet.
- The sender maintains a **window size** of one, which means it can have

only one unacknowledged packet in the network at any time.

- The receiver sends an acknowledgment for each packet it receives, and discards any duplicate packets.
- The sender uses a **timer** to detect packet losses or delays. If the timer expires before receiving an acknowledgment, the sender retransmits the packet.
- Stop-and-wait is simple and easy to implement, but it has low efficiency and throughput, especially for long-distance or high-speed networks, because the sender has to wait for a round-trip time (RTT) between each packet transmission.

Sliding window

- In sliding window, the sender can send multiple packets without waiting for acknowledgments, as long as the number of unacknowledged packets does not exceed the window size.
- The sender maintains a **sending window**, which is a range of sequence numbers of packets that can be sent at any time. The sender also keeps track of the **last acknowledgment received (LAR)** and the **last packet sent (LPS)**.
- The receiver maintains a **receiving window**, which is a range of sequence numbers of packets that can be received and acknowledged at any time. The receiver also keeps track of the **last packet received (LPR)** and the **largest acceptable packet (LAP)**.
- The receiver sends a **cumulative acknowledgment** for the last packet it received in order, and also indicates the size of its receiving window. The receiver can also use **selective acknowledgments** to inform the sender of specific packets that have been received out of order.
- The sender uses a timer to detect packet losses or delays. If the timer expires for a packet, the sender retransmits the packet and all subsequent packets in the sending window.
- Sliding window can be further classified into two types: **go-back-N** and **selective repeat**.

Go-back-N

- In go-back-N, the sender retransmits all the packets in the sending window when a packet loss or delay is detected.
- The sender uses a fixed window size of N, which is usually equal to the size of the receiver's window.
- The receiver discards any out-of-order packets and sends a cumulative acknowledgment for the last packet received in order.
- Go-back-N is simple and easy to implement, but it has low efficiency and reliability, especially for networks with high error rates, because it wastes bandwidth and time by retransmitting packets that have already been received by the receiver.

Selective repeat

- In selective repeat, the sender retransmits only the packets that have been lost or delayed when a packet loss or delay is detected.
- The sender uses a variable window size, which is usually smaller than the size of the receiver's window, to avoid sending too many packets at once.
- The receiver buffers any out-of-order packets and sends a selective acknowledgment for each packet received, indicating the sequence number and the size of its receiving window.
- Selective repeat is more complex and difficult to implement, but it has high efficiency and reliability, especially for networks with high error rates, because it avoids retransmitting packets that have already been received by the receiver.

TCP Congestion control in transport layer

- TCP stands for Transmission Control Protocol. It is a connection-oriented and reliable transport layer protocol that facilitates the transmission of packets from source to destination.
- TCP congestion control is the main technique that prevents the Internet from slowing down due to over-utilization of network resources.
- TCP congestion control aims to achieve two goals: high throughput and fairness. High throughput means that the sender can utilize the available bandwidth efficiently and avoid wasting network resources. Fairness means that the sender can share the network resources fairly with other competing flows and avoid starving them.
- TCP congestion control is based on the concept of congestion window (cwnd), which is the amount of data that the sender can send without receiving an acknowledgment from the receiver. The sender adjusts the cwnd based on the feedback from the network, such as acknowledgments, timeouts, and congestion signals.
- There are three phases that TCP uses for congestion control: slow start, congestion avoidance, and congestion detection:
 - Slow start: In this phase, the sender starts with a small cwnd and doubles it every round trip time (RTT) until it reaches a threshold (ssthresh) or a loss occurs. This phase allows the sender to probe the network capacity and find the optimal cwnd.
 - Congestion avoidance: In this phase, the sender increases the cwnd by one segment every RTT, which results in a linear growth of cwnd. This phase allows the sender to maintain a high throughput and avoid causing congestion.
 - Congestion detection: In this phase, the sender detects a loss event, which indicates that the network is congested. The sender reduces the cwnd and the ssthresh by half and enters either slow start or congestion avoidance phase depending on the new cwnd value. This phase allows the sender to react to congestion and reduce the network

load.

- TCP uses different mechanisms to detect congestion, such as timeouts, duplicate acknowledgments, and explicit congestion notification (ECN):
 - Timeouts: A timeout occurs when the sender does not receive an acknowledgment for a segment within a certain time interval. This indicates that the segment was lost or delayed in the network. The sender retransmits the segment and enters the slow start phase.
 - Duplicate acknowledgments: A duplicate acknowledgment occurs when the receiver receives an out-of-order segment and sends an acknowledgment for the last in-order segment. This indicates that a segment was lost or reordered in the network. If the sender receives three duplicate acknowledgments for the same segment, it performs a fast retransmit and enters the congestion avoidance phase. This is also known as fast recovery.
 - Explicit congestion notification (ECN): ECN is a mechanism that allows the network devices, such as routers and switches, to mark the packets with a congestion signal when they experience congestion. The receiver echoes the congestion signal back to the sender in the acknowledgment. The sender reduces the cwnd and enters the congestion avoidance phase. ECN is not widely used and most operating systems have it disabled by default.
- TCP congestion control is a dynamic and adaptive process that adjusts the cwnd according to the network conditions. It is a key factor that affects the performance and efficiency of TCP and the Internet.

Quality of service in transport layer

- Quality of service (QoS) refers to the ability of a network to provide different levels of service to different types of traffic, such as voice, video, or data.
- QoS in transport layer is mainly concerned with the end-to-end performance of a communication session, such as delay, jitter, bandwidth, reliability, and error rate.
- QoS in transport layer can be achieved by using different protocols, mechanisms, and techniques, such as:
 - TCP and UDP: TCP provides reliable, ordered, and error-free delivery of data, but may incur high overhead and delay. UDP provides fast and low-overhead delivery of data, but does not guarantee reliability, order, or error correction. Depending on the application requirements, either TCP or UDP can be used to provide different QoS levels.
 - Congestion control and flow control: Congestion control is the process of adjusting the sending rate of data packets in response to network congestion, to avoid packet loss and improve network utilization. Flow control is the process of regulating the amount of data that a sender can transmit to a receiver, to avoid overwhelming the

receiver's buffer. Both congestion control and flow control can help to improve QoS by preventing packet loss, delay, and jitter.

- Stream Control Transmission Protocol (SCTP): SCTP is a transport layer protocol that provides reliable, ordered, and error-free delivery of data, similar to TCP. However, SCTP also supports multiple streams within a single connection, which can improve QoS by allowing different types of data to be transmitted in parallel, without blocking or reordering. SCTP also supports multihoming, which can improve QoS by allowing a connection to switch between multiple network paths, in case of failure or congestion.
- Real-time Transport Protocol (RTP): RTP is a transport layer protocol that is designed for real-time applications, such as voice and video. RTP provides features such as timestamping, sequencing, and payload identification, which can help to synchronize and identify the data packets. RTP also supports QoS feedback mechanisms, such as Real-time Transport Control Protocol (RTCP), which can provide information about the network conditions, such as packet loss, delay, and jitter. RTP can be used with UDP or SCTP to provide different QoS levels for real-time applications.
- Differentiated Services (DiffServ): DiffServ is a network architecture that provides QoS by classifying and marking the data packets according to their priority and service level. DiffServ uses a 6-bit field in the IP header, called the Differentiated Services Code Point (DSCP), to indicate the QoS level of each packet. DiffServ routers can then apply different policies and treatments to the packets based on their DSCP values, such as queuing, scheduling, shaping, or dropping. DiffServ can provide QoS in transport layer by allowing the applications to specify their QoS requirements and preferences, and by enabling the network to deliver the packets accordingly.

Unit 5 - Application Layer in Computer Networks

- The application layer is the highest layer of the Open Systems Interconnection (OSI) model and the Internet Protocol Suite (TCP/IP) model. It is where users interact with the network, download information and send data.
- The application layer is not an application itself, but an abstraction layer that specifies the shared communications protocols and interface methods used by hosts in a communications network. It ensures that an application can effectively communicate with other applications on different computer systems and networks.
- The application layer provides various services and functions to the users and applications, such as:
 - Email: The application layer provides a facility by which users can forward several emails and it also provides a storage facility. Some of the protocols used for email are Simple Mail Transfer Protocol

(SMTP), Post Office Protocol (POP) and Internet Message Access Protocol (IMAP).

- File Transfer: The application layer allows users to access, retrieve and manage files in a remote computer. Some of the protocols used for file transfer are File Transfer Protocol (FTP), Trivial File Transfer Protocol (TFTP) and Secure File Transfer Protocol (SFTP).
- Remote Login: The application layer allows users to log on as a remote host. Some of the protocols used for remote login are Telnet, Secure Shell (SSH) and Remote Desktop Protocol (RDP).
- Directory Services: The application layer provides access to global information about various objects and services on the network. Some of the protocols used for directory services are Lightweight Directory Access Protocol (LDAP), Domain Name System (DNS) and Active Directory.
- Web Browsing: The application layer enables users to access and view web pages on the internet. Some of the protocols used for web browsing are Hypertext Transfer Protocol (HTTP), Hypertext Transfer Protocol Secure (HTTPS) and HyperText Markup Language (HTML).
- Multimedia: The application layer supports various types of multimedia applications, such as voice over IP, video conferencing, streaming and online gaming. Some of the protocols used for multimedia are Real-time Transport Protocol (RTP), Real-time Transport Control Protocol (RTCP), Session Initiation Protocol (SIP) and H.323.
- Network Management: The application layer provides tools and functions for monitoring and controlling the network devices and resources. Some of the protocols used for network management are Simple Network Management Protocol (SNMP), Common Management Information Protocol (CMIP) and Network Time Protocol (NTP).
- The application layer protocols are classified into two types: specific and generic. Specific protocols are designed for a particular application, such as HTTP for web browsing and SMTP for email. Generic protocols are designed to support multiple applications, such as TCP and UDP for data transmission and IP for addressing and routing.
- The application layer protocols use different methods of communication, such as request-response, client-server, peer-to-peer and publish-subscribe. Request-response is a method where one party sends a request and the other party sends a response. Client-server is a method where one party acts as a client and requests a service from another party that acts as a server. Peer-to-peer is a method where both parties act as peers and exchange data directly without a central server. Publish-subscribe is a method where one party publishes a message and the other party subscribes to receive the message.
- The application layer protocols use different formats of data representation, such as text, binary, XML and JSON. Text is a format where data is represented as human-readable characters. Binary is a format where data

is represented as bits and bytes. XML is a format where data is represented as tags and attributes. JSON is a format where data is represented as key-value pairs.

- The application layer protocols use different methods of data compression, encryption and authentication to reduce the size, protect the confidentiality and verify the identity of the data. Data compression is a method where data is reduced by removing redundant or irrelevant information. Data encryption is a method where data is transformed by using a secret key or algorithm. Data authentication is a method where data is verified by using a password, certificate or token.

Domain Name System

- The Domain Name System (DNS) is a hierarchical and distributed naming system for computers, services, and other resources in the Internet or other Internet Protocol (IP) networks.
- DNS is the main index of the internet that directs traffic for queries across the web. It maps the name people use to locate a website to the IP address that a computer uses to locate that website.
- DNS is an internet service that translates the domain name into IP addresses. For example, the domain name `google.com` is translated to the IP address `142.250.72.238`.
- DNS works by using a network of servers called DNS servers. When a user requests a domain name, the request first goes to a DNS server. The DNS server then looks up the domain name in its database and returns the corresponding IP address to the user. If the DNS server does not have the domain name in its database, it forwards the request to another DNS server until the IP address is found or the request fails.
- DNS has a hierarchical structure, which means that there are different levels of DNS servers. The top level is the root DNS servers, which store the information about the top-level domains (TLDs), such as `.com`, `.org`, `.net`, etc. The second level is the authoritative DNS servers, which store the information about the specific domains within each TLD, such as `google.com`, `okta.com`, `cloudflare.com`, etc. The third level is the local DNS servers, which are usually provided by the internet service providers (ISPs) or the organizations, and cache the information from the authoritative DNS servers for faster access.
- DNS uses a protocol called DNS resolution to find the IP address of a domain name. DNS resolution involves sending a query message from the user to the DNS server, and receiving a response message from the DNS server to the user. The query message contains the domain name that the user wants to resolve, and the response message contains the IP address of the domain name or an error code if the resolution fails.

- DNS resolution can be iterative or recursive. In iterative resolution, the DNS server returns the best answer it has, or a referral to another DNS server that may have the answer. The user then sends the query to the referred DNS server and repeats the process until the IP address is found or the query fails. In recursive resolution, the DNS server does the work of contacting other DNS servers on behalf of the user, and returns the final answer or an error to the user. Recursive resolution is faster and more convenient for the user, but it puts more load on the DNS server.
- DNS has many benefits, such as:
 - It makes the internet more user-friendly, as people can use easy-to-remember domain names instead of complex IP addresses.
 - It enables the scalability and flexibility of the internet, as domain names can be changed or moved to different IP addresses without affecting the users.
 - It supports the security and reliability of the internet, as DNS servers can implement various mechanisms to prevent or mitigate attacks, such as DNS spoofing, DNS hijacking, DNS cache poisoning, etc.
- DNS also has some challenges, such as:
 - It can be vulnerable to various attacks, as malicious actors can exploit the weaknesses of the DNS protocol or the DNS servers to redirect or block the users from accessing the legitimate websites.
 - It can be affected by various factors, such as network congestion, server overload, configuration errors, etc., which can cause delays or failures in the DNS resolution process.
 - It can be inconsistent or outdated, as DNS servers may have different or old information about the domain names and their IP addresses, which can lead to errors or conflicts in the DNS resolution process.
- A possible mnemonic to remember the steps of DNS resolution is:
 - **D**omain name requested by user
 - **N**ame server queried by user
 - **S**erver responds with IP address or referral
 - **R**epeat until IP address found or query fails
 - **E**nd of resolution process
 - **S**uccess or error returned to user
 - **O**utput displayed to user
 - **L**ocal cache updated by user
- A possible learning trick to understand the hierarchical structure of DNS is to compare it to a family tree, where:
 - The root DNS servers are the ancestors, who know the information about the main branches of the family (TLDs).
 - The authoritative DNS servers are the parents, who know the information about their

World Wide Web

- The World Wide Web (WWW) is an information system that enables documents and other web resources to be accessed over the Internet.
- The WWW is not the same as the Internet, which is a global network of computers that can communicate with each other. The WWW is a service that operates over the Internet, just as email and Usenet do.
- The WWW consists of webpages that are linked by hyperlinks, which are references to other webpages or web resources. Users can navigate the WWW by clicking on hyperlinks or typing in web addresses (URLs) in web browsers.
- The WWW was invented by Tim Berners-Lee in 1989 at CERN, the European Organization for Nuclear Research. He proposed a system of hypertext documents that could be viewed by anyone on the Internet using a standard protocol called HTTP.
- The WWW gained rapid acceptance with the creation of a web browser called Mosaic, which was developed by Marc Andreessen and others at the National Center for Supercomputing Applications at the University of Illinois and was released in September 1993. Mosaic was the first graphical web browser that could display images and multimedia content along with text.
- The WWW has revolutionized the fields of communication, education, entertainment, commerce, and social networking. It has also raised issues of privacy, security, censorship, and digital divide.

Some mnemonics and learning tricks for the WWW are:

- WWW stands for World Wide Web, which can be remembered as W3 or 3W.
- WWW is a service that operates over the Internet, which can be remembered as WWW is a subset of the Internet, or WWW is inside the Internet.
- WWW consists of webpages that are linked by hyperlinks, which can be remembered as WWW is a web of links, or WWW is a network of pages.
- WWW was invented by Tim Berners-Lee, which can be remembered as WWW is TBL's baby, or WWW has three letters and TBL has three initials.
- WWW gained popularity with Mosaic, which can be remembered as WWW and Mosaic both start with M, or Mosaic made WWW more colorful.

Hyper Text Transfer Protocol

- Hyper Text Transfer Protocol (HTTP) is an application-layer protocol for transmitting hypermedia documents, such as HTML .
- HTTP is the underlying protocol used by the World Wide Web, developed by Tim Berners-Lee .
- HTTP defines how messages are formatted and transmitted, and what

actions Web servers and browsers should take in response to various commands .

- HTTP is a stateless protocol, which means that each request is independent of the previous ones and the server does not keep track of the client's state .
- HTTP uses a client-server model, where the client initiates a request and the server responds with a response .
- HTTP requests and responses consist of a start-line, zero or more headers, an empty line, and an optional message body .
- The start-line of a request contains the method, the request-target, and the HTTP version . For example:

```
GET /index.html HTTP/1.1
```

- The start-line of a response contains the HTTP version, the status code, and the reason phrase . For example:

```
HTTP/1.1 200 OK
```

- The headers provide additional information about the request or the response, such as the content type, the content length, the date, the server name, etc . For example:

```
Content-Type: text/html
Content-Length: 1234
Date: Mon, 13 Mar 2023 12:48:03 GMT
Server: Apache
```

- The message body contains the actual data that is being transferred, such as the HTML document, the image, the JSON object, etc . For example:

```
<html>
<head>
<title>Example</title>
</head>
<body>
<h1>Hello, world!</h1>
</body>
</html>
```

- HTTP supports different methods for different purposes, such as GET, POST, PUT, DELETE, etc .
- GET is used to retrieve a resource from the server, such as a web page or an image .
- POST is used to send data to the server, such as a form submission or a file upload .
- PUT is used to update or create a resource on the server, such as a new document or a modified image .
- DELETE is used to delete a resource from the server, such as a document or an image .

- HTTP supports different status codes to indicate the outcome of a request, such as 200, 404, 500, etc .
- 200 means OK, which means that the request was successful and the response contains the requested resource .
- 404 means Not Found, which means that the request was unsuccessful and the server could not find the requested resource .
- 500 means Internal Server Error, which means that the request was unsuccessful and the server encountered an unexpected error .
- HTTP can be secured by using HTTPS, which is HTTP over TLS (Transport Layer Security) .
- HTTPS encrypts the communication between the client and the server, preventing eavesdropping, tampering, and impersonation .
- HTTPS uses certificates to verify the identity of the server and the client, ensuring that they are who they claim to be .
- HTTPS requires a valid certificate from a trusted authority, such as Let's Encrypt, VeriSign, or Comodo .

Some mnemonics and learning tricks for HTTP are:

- HTTP stands for Hyper Text Transfer Protocol, which can be remembered as **H**ow **T**o **T**ransfer **P**ages.
- HTTP methods can be remembered as **G**et **P**ost **P**ut **D**ele~~t~~e, which can be pronounced as **GiPPy Doo**.
-

Electronic mail in application layer

- Electronic mail (or email) is an application layer service that allows users to exchange messages and information over the internet .
- Email is one of the most popular and widely used services of the internet.
- Email has three main components: user agents, mail servers, and protocols .
 - User agents are the applications that allow users to create, read, and send emails. Examples of user agents are Outlook, Gmail, Thunderbird, etc .
 - Mail servers are the computers that store and forward emails. Each mail server has a mailbox for each user and a message queue for outgoing emails .
 - Protocols are the rules and standards that govern the communication between user agents and mail servers. There are two main types of protocols: SMTP and POP/IMAP .
- SMTP (Simple Mail Transfer Protocol) is the protocol that is used to send emails from one mail server to another. SMTP uses a client-server model,

where the sender's mail server acts as the client and the receiver's mail server acts as the server .

- SMTP has three phases: handshaking, transfer, and closure .
 - Handshaking is the phase where the client and the server establish a connection and exchange greetings and parameters .
 - Transfer is the phase where the client sends the email message to the server, along with the sender and receiver addresses .
 - Closure is the phase where the client and the server terminate the connection and acknowledge the successful or unsuccessful transfer .
- SMTP uses a simple text-based format for the email message, which consists of two parts: the header and the body .
 - The header contains information such as the sender, receiver, subject, date, etc .
 - The body contains the actual content of the message, which can be plain text, HTML, or attachments .
- SMTP can only handle ASCII characters, so if the message contains non-ASCII characters or binary data, it has to be encoded using schemes such as MIME (Multipurpose Internet Mail Extensions) or Base64 .
- SMTP does not provide any security or authentication features, so it is vulnerable to attacks such as spamming, spoofing, or interception .
- POP (Post Office Protocol) and IMAP (Internet Message Access Protocol) are the protocols that are used to retrieve emails from the mail server to the user agent. POP and IMAP use a client-server model, where the user agent acts as the client and the mail server acts as the server .
- POP allows the user to download the emails from the mail server and delete them from the server. POP is simple and efficient, but it does not support multiple clients or synchronization .
- IMAP allows the user to access the emails on the mail server without downloading them. IMAP supports multiple clients, synchronization, and folder management. IMAP is more complex and flexible, but it requires more bandwidth and storage .
- POP and IMAP also do not provide any security or authentication features, so they are vulnerable to attacks such as eavesdropping, tampering, or impersonation .
- To enhance the security and authentication of email protocols, extensions such as SSL (Secure Sockets Layer), TLS (Transport Layer Security), or S/MIME (Secure/Multipurpose Internet Mail Extensions) can be used .
- A possible mnemonic to remember the email protocols is: **Send Messages To People, Pick Or Peek, Inbox M**

File Transfer Protocol in application layer

- File Transfer Protocol (FTP) is an application layer protocol that allows users to transfer files between hosts over a TCP/IP network.
- FTP uses two TCP connections: a control connection and a data connection.
- The control connection is used to exchange commands and responses between the client and the server. The data connection is used to transfer the actual files.
- FTP follows a client-server model, where the client initiates the connection and requests files from the server. The server responds to the client's requests and sends the files.
- FTP supports two modes of operation: active mode and passive mode.
- In active mode, the client opens a random port for the data connection and sends its port number to the server through the control connection. The server then connects to the client's port to send the files.
- In passive mode, the server opens a random port for the data connection and sends its port number to the client through the control connection. The client then connects to the server's port to receive the files.
- Passive mode is preferred over active mode because it avoids the problem of firewalls blocking incoming connections to the client.
- FTP supports two types of file transfer: binary and ASCII.
- Binary transfer is used to transfer files that are not text-based, such as images, audio, video, etc. Binary transfer preserves the exact format and content of the files.
- ASCII transfer is used to transfer text-based files, such as HTML, XML, etc. ASCII transfer converts the files to a standard text format that can be read by different systems.
- FTP supports various commands and responses to perform different operations, such as:
 - USER and PASS: to authenticate the user and password.
 - LIST and NLST: to list the files and directories in the current directory.
 - CWD and CDUP: to change the current directory and move up one level.
 - RETR and STOR: to retrieve and store files from and to the server.
 - PORT and PASV: to switch between active and passive mode.
 - TYPE: to specify the type of file transfer (binary or ASCII).
 - QUIT: to terminate the FTP session.
- FTP uses the following format for commands and responses:
 - Commands: `<command> <argument>\r\n`
 - Responses: `<status code> <message>\r\n`
- FTP uses three-digit status codes to indicate the status of the commands and responses, such as:
 - 1xx: positive preliminary reply, indicating that the command is accepted but more information is needed.

- 2xx: positive completion reply, indicating that the command is successfully completed.
- 3xx: positive intermediate reply, indicating that the command is accepted but another command is needed.
- 4xx: transient negative reply, indicating that the command is not accepted due to a temporary error.
- 5xx: permanent negative reply, indicating that the command is not accepted due to a permanent error.
- FTP is widely used for transferring files over the Internet, such as downloading software, uploading web pages, etc.
- FTP has some advantages and disadvantages, such as:
 - Advantages:
 - * FTP is simple and easy to use.
 - * FTP can transfer large files efficiently and reliably.
 - * FTP can resume interrupted transfers.
 - * FTP can support different types of file transfer.
 - Disadvantages:
 - * FTP is not secure, as it sends the user credentials and data in plain text.
 - * FTP does not support encryption or compression of the data.
 - * FTP does not support file synchronization or version control.
 - * FTP does not support concurrent access or locking of the files.

Remote login in application layer

- Remote login is a process in which a user can log in to a remote computer and use the services that are available on that computer.
- Remote login is an example of an application layer service, which is the highest layer in the OSI model and the TCP/IP model of computer networking.
- The application layer provides the interface between the user and the network, and defines the protocols for various applications such as file transfer, email, web browsing, etc.
- Remote login is implemented by protocols such as Telnet, SSH, Rlogin, etc. that allow a user to establish a terminal session with a remote host over a network.
- The basic steps involved in remote login are:
 - The user initiates a connection request to the remote host using a client program (such as Telnet or SSH) on the local computer.
 - The remote host responds to the connection request and authenticates the user using a username and password or other methods.
 - If the authentication is successful, the remote host allocates a pseudo-terminal for the user and executes a shell program that allows the user to interact with the remote system.
 - The user can then execute commands, access files, run programs, etc. on the remote system as if they were physically present at the remote

location.

- The user can terminate the session by typing exit or logout on the shell prompt or by closing the client program on the local computer.
- The advantages of remote login are:
 - It allows the user to access and use the resources of a remote computer without physically moving to that location.
 - It enables the user to perform tasks that require higher processing power, storage capacity, or specialized software that are not available on the local computer.
 - It enhances the security and privacy of the user by encrypting the data transmitted over the network and preventing unauthorized access to the local computer.
- The disadvantages of remote login are:
 - It depends on the availability and reliability of the network connection between the local and remote computers.
 - It may cause network congestion and latency due to the large amount of data transferred over the network.
 - It may expose the user to potential security risks such as malware, phishing, or man-in-the-middle attacks if the remote host or the network is compromised.
- A mnemonic to remember the steps involved in remote login is: **CARTER** (Connect, Authenticate, Run, Terminate, Encrypt, Repeat).
- An example of remote login is using SSH to connect to a Linux server from a Windows laptop and running a Python script on the server.

Network management in application layer

- Network management is the process of configuring, monitoring, and managing the performance of a network.
- The application layer is the topmost layer in the Open System Interconnection (OSI) model and the Internet Protocol Suite (TCP/IP) model.
- The application layer provides several ways for manipulating the data (information) which actually enables any type of user to access network with ease.
- The application layer also handles network services, such as file and printing, name resolution, and redirector services.
- Name resolution is the process of mapping an IP address to a human-readable name, such as a domain name.
- Redirector services allow a user to access resources on another network device, such as a file server or a printer.
- Some examples of application layer protocols are HTTP, FTP, SMTP, DNS, SNMP, and Telnet.
- HTTP (Hypertext Transfer Protocol) is used for transferring web pages and other resources over the internet.
- FTP (File Transfer Protocol) is used for transferring files between hosts

on a network.

- SMTP (Simple Mail Transfer Protocol) is used for sending and receiving email messages.
- DNS (Domain Name System) is used for resolving domain names to IP addresses and vice versa.
- SNMP (Simple Network Management Protocol) is used for collecting and organizing information about network devices and their status.
- Telnet is used for providing remote access to a host on a network.
- Network management systems use application layer protocols, such as SNMP, to communicate with network devices and collect information about their configuration, performance, and faults.
- A network management system consists of three components: a manager, an agent, and a management information base (MIB).
- A manager is a software application that runs on a network management platform and sends requests to agents on network devices.
- An agent is a software component that runs on a network device and responds to requests from managers.
- A MIB is a database that stores information about the network device and its capabilities, such as its name, location, type, status, and configuration.
- A network management system can perform various functions, such as fault management, performance management, configuration management, security management, and accounting management.
- Fault management is the process of detecting, isolating, and correcting network problems.
- Performance management is the process of measuring and analyzing network performance and ensuring that it meets the service level agreements (SLAs).
- Configuration management is the process of maintaining and updating the configuration of network devices and ensuring that they are consistent and compliant with the policies.
- Security management is the process of protecting the network from unauthorized access and malicious attacks.
- Accounting management is the process of collecting and reporting network usage and billing information.
- A network management system can use various tools and techniques, such as polling, traps, syslog, thresholds, alarms, reports, and dashboards, to perform these functions.
- Polling is the technique of periodically sending requests to agents on network devices and receiving responses with the current status and performance data.
- Traps are the technique of sending unsolicited messages from agents to managers when a significant event or a fault occurs on a network device.
- Syslog is the technique of sending log messages from network devices to a central server for analysis and troubleshooting.
- Thresholds are the technique of setting predefined values for network performance metrics and comparing them with the actual values.

- Alarms are the technique of generating notifications when a threshold is crossed or a fault is detected.
- Reports are the technique of summarizing and presenting network performance and usage data in a readable format.
- Dashboards are the technique of displaying network performance and status data in a graphical and interactive way.
- A network management system can benefit the network administrators and users in various ways, such as improving network availability, reliability, efficiency, security, and scalability.
- Network availability is the measure of how often the network is operational and accessible.
- Network reliability is the measure of how well the network can handle failures and recover from them.
- Network efficiency is the measure of how well the network can utilize its resources and deliver the required service quality.

Data compression in application layer

- Data compression is the process of reducing the size of data by applying some encoding techniques that eliminate redundancy and irrelevant information.
- Data compression can be performed at different layers of the network stack, but the application layer is the most common place where compression is applied.
- Data compression at the application layer can be classified into two types: lossless and lossy.
- Lossless compression preserves the exact information of the original data, while lossy compression discards some information that is deemed less important or perceptually irrelevant.
- Lossless compression is suitable for text, code, and other data that require exact reproduction, while lossy compression is suitable for images, audio, video, and other data that can tolerate some degradation in quality.
- Some examples of lossless compression algorithms are Huffman coding, Lempel-Ziv-Welch (LZW) coding, run-length encoding (RLE), and deflate.
- Some examples of lossy compression algorithms are JPEG, MPEG, MP3, and AAC.
- Data compression at the application layer has several advantages, such as:
 - Reducing the bandwidth consumption and transmission time of data over the network.
 - Saving the storage space and cost of data on the devices.
 - Enhancing the user experience and quality of service of the applications.
- Data compression at the application layer also has some disadvantages, such as:
 - Increasing the computational complexity and overhead of the compression and decompression processes.

- Introducing some errors and artifacts in the compressed data, especially for lossy compression.
- Requiring some compatibility and standardization of the compression formats and protocols among the applications and devices.

Cryptography in application layer

- Cryptography in application layer is the process of encrypting and decrypting data at the level of the application that is sending or receiving the data.
- Cryptography in application layer can be used to protect any type of data that passes through an application, such as text, images, audio, video, etc.
- Cryptography in application layer can increase the security of data by encrypting it across multiple layers, such as disk, file, and database, and reducing the number of potential attack vectors .
- Cryptography in application layer can also provide end-to-end encryption, which means that only the sender and the receiver of the data have the keys to decrypt it, and no intermediate party can access the data.
- Cryptography in application layer can be implemented using various algorithms and techniques, such as symmetric encryption, asymmetric encryption, hashing, digital signatures, etc.
- Cryptography in application layer can be applied to different types of applications, such as native apps, browser apps, web apps, mobile apps, etc.
- Cryptography in application layer can have various advantages and disadvantages, depending on the context and the requirements of the application.

Some of the advantages are:

- It can provide data confidentiality, integrity, and authenticity.
- It can prevent unauthorized access, modification, or deletion of data.
- It can enable secure communication and data exchange between parties.
- It can comply with data protection regulations and standards.

Some of the disadvantages are:

- It can increase the complexity and overhead of the application development and maintenance.
- It can introduce performance and latency issues for the application and the network.
- It can require careful key management and distribution policies and practices.
- It can be vulnerable to cryptographic attacks or human errors.

Some of the examples of cryptography in application layer are:

- Secure Sockets Layer (SSL) and Transport Layer Security (TLS) protocols, which provide encryption and authentication for web applications and

browsers.

- Pretty Good Privacy (PGP) and GNU Privacy Guard (GPG) software, which provide encryption and digital signatures for email and file applications.
- Signal and WhatsApp applications, which provide end-to-end encryption for instant messaging and voice calls.
- Bitcoin and Ethereum applications, which use cryptography for creating and verifying transactions and blocks in the blockchain network.

Basic concepts of Cryptography in application layer

Cryptography is the science of securing communications from unauthorized parties. It involves the use of mathematical techniques to transform data into unintelligible forms and back. Cryptography can provide confidentiality, integrity, authenticity, and non-repudiation to the data transmitted or stored in applications.

Some of the basic concepts of cryptography in the application layer are:

- **Plain text** and **cipher text**: Plain text is the original data or message that needs to be protected. Cipher text is the encrypted data or message that is unreadable by anyone without the proper key. An example of plain text is “Hello, world!” and an example of cipher text is “Uryyb, jbeyq!” (using ROT13 encryption).
- **Algorithm** and **key**: An algorithm is a set of rules or steps that defines how to encrypt or decrypt data. A key is a secret value that is used as an input to the algorithm. The key determines the output of the algorithm and must be known by both the sender and the receiver of the encrypted data. An example of an algorithm is AES (Advanced Encryption Standard) and an example of a key is “1234567890abcdef”.
- **Encryption** and **decryption**: Encryption is the process of transforming plain text into cipher text using an algorithm and a key. Decryption is the reverse process of transforming cipher text into plain text using the same or a different algorithm and key. An example of encryption is using AES and “1234567890abcdef” to encrypt “Hello, world!” into “3q2+7w4w9v/6WsQ6Liw=”
- **Symmetric** and **asymmetric** cryptography: Symmetric cryptography is a type of cryptography that uses the same key for both encryption and decryption. It is fast and efficient, but it requires a secure way of sharing the key between the sender and the receiver. An example of symmetric cryptography is AES. Asymmetric cryptography, also known as public key cryptography, is a type of cryptography that uses a pair of keys for encryption and decryption: a public key and a private key. The public key can be shared with anyone, but the private key must be kept secret. The public key can be used to encrypt data, but only the private key can decrypt it, and vice versa. It is more secure and flexible, but it is slower and more complex than symmetric cryptography. An example of

asymmetric cryptography is RSA (Rivest-Shamir-Adleman).

- **Digital signature and digital certificate:** A digital signature is a way of proving the authenticity and integrity of a message or a document. It is created by encrypting a hash (a fixed-length summary) of the message or document with the sender's private key. The receiver can verify the signature by decrypting it with the sender's public key and comparing the hash with the original message or document. A digital certificate is a way of proving the identity and validity of a public key. It is issued by a trusted authority (a certificate authority or CA) and contains information such as the public key, the owner's name, the expiration date, and the CA's signature. An example of a digital signature is using RSA and SHA-256 (Secure Hash Algorithm) to sign a message. An example of a digital certificate is using X.509 (a standard format for certificates) to certify a public key.